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BSD4543 DEEP LEARNING

TITLE:

WHEEL PROFILING AND REPLACEMENT FORECAST WITH MEAN TIME BEFORE
FAILURE (MTBF) PREDICTION FOR MRT KAJANG LINE

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CHAPTER 1

INTRODUCTION

1.1 Research Background

The four strategic missions of Malaysia's New Industrial Master Plan 2030 (NIMP 2030) are to advancing economic complexity, tech up for a digitally vibrant nation, pushing for net zero and safeguarding economic security and inclusivity. Among these, the goals of "Tech Up for a Digitally Vibrant Nation" is closely related to this project, which focuses on implementing predictive modelling and advanced data analytics into the Mass Rapid Transit (MRT) Kajang Line's maintenance processes. This project supports national initiatives to digitally transform traditional industries, improve operations, and encourage high-value technological capabilities in public infrastructure management by using forecasting techniques and artificial intelligence to operational railway data.

This project also covers company-specific issues in the transportation sector, specifically how data analytics might enhance MRT train wheel maintenance's speed, accuracy, and efficiency. Predicting failures before they happen enables data-driven decision-making, which is crucial for preventing unexpected malfunctions, enhancing passenger safety, and decreasing maintenance costs. This is particularly crucial for the MRT Kajang Line, which operates a fleet of 58 trains. Figure 1 shows the structure of each train that is composed of four cars (Motor Car 1 (M1), Trailer Car (T1), Trailer Car (T2), Motor Car 2 (M2)). The difference between the motor car and trailer car is that motor car has motor, but the trailer car does not have motor as shown in the figures 2 and 3. Each car is equipped with two bogies. Each bogie has two wheelsets (axle and wheel) as shown in the figure 4.

M1	T1	T2	M2
----	----	----	----

Figure 1: Each train composed of four cars.



Figure 2: Motor Car

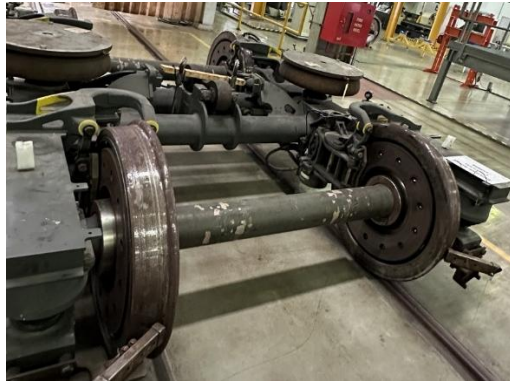


Figure 3: Trailer Car

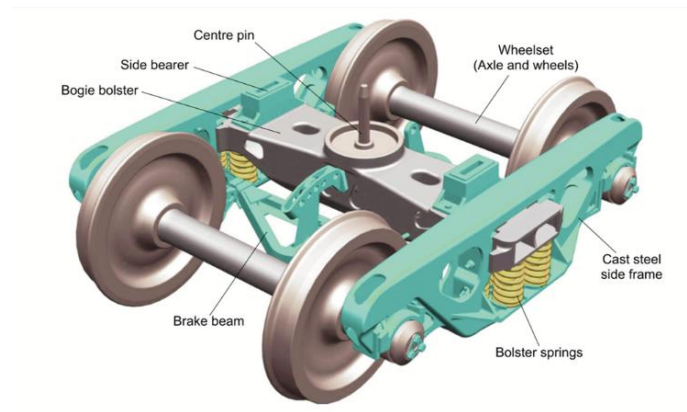


Figure 4: Bogie of train (PRC Rail Consulting Ltd, 2023).

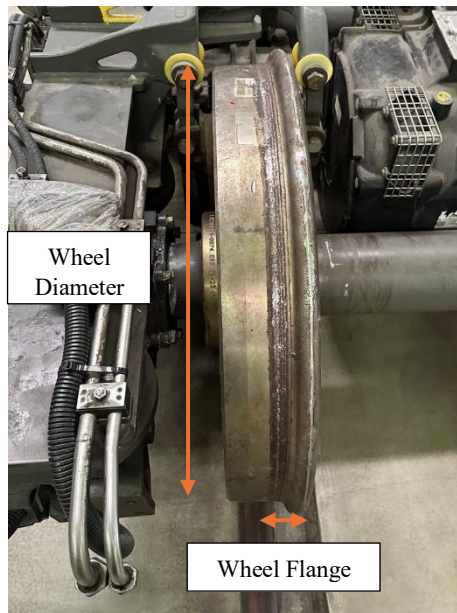


Figure 5: Wheel diameter and flange of train.

For this project, eight trains were chosen out of 58 trains which T01, T05, T25, T45, T46, T55, T57, and T58. These were chosen based on a combination of different maintenance profiles and operational histories. A typical wheelset consists of the wheel, axle, and wheel

flange with particular focus on to the P8 standard design specification which determines acceptable limits and wear tolerances for wheel geometry. Wheel replacement is required when wear exceeds critical thresholds and wheel profiling is a machining process that restores the wheel's original structure. According to Ye et al. (2021), an appropriate railway wheel profile can increase the dynamic performance of trains on tracks, increase the wheelsets' lifespan, or save maintenance costs. An example of a measured wheel profile plotted with the original wheel profile's shape is shown in Figure 6 (Asplund et al., 2014). This two are important when do maintenance tasks. In this situation, failure is defined as a wheel flange thickness less than 27.5 mm or wheel diameter less than 775 mm or both of it. Wheel flange thickness and wheel diameter shown in the figure 5. These values are used in the Mean Time Before Failure (MTBF), a crucial indicator for predicting how long each wheel component will stay useful.

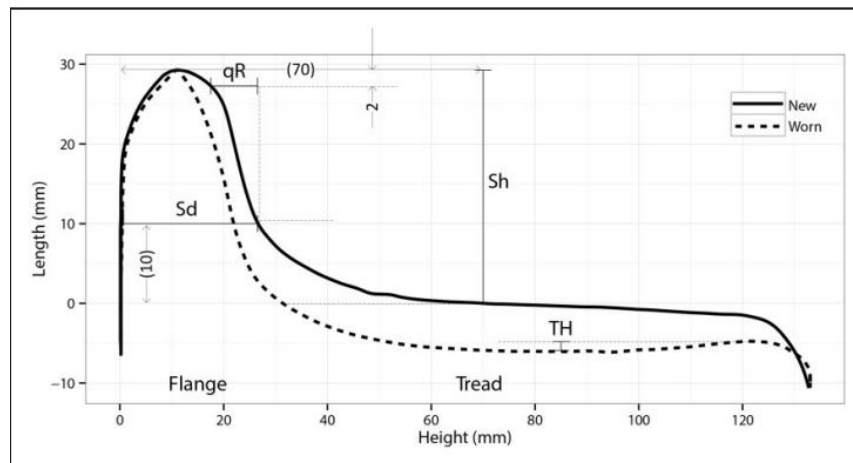


Figure 6: Original and worn wheel profiles measured.

Furthermore, this project used three model to solve the problem which is Linear Discriminant Analysis (LDA), Holt-Winters Method and Gated Recurrent Unit (GRU) model. A traditional model recognition algorithm is LDA also known as Fisher Liner Discriminant (Feng Han et al., 2023). By using LDA, we can identify the most relevant variables that differentiate between failure and no failure situations. The framework for forecasting models that try to forecast when wheel profiling and replacement is likely to be required is provided by this feature selection. To do this, two methods are used which is Holt-Winters Method and GRU. Holt-Winters method also known as Triple Exponential Smoothing. According to (Raden Gunawan Santosa et al., 2024), the Holt-Winter method was developed especially for data with regular seasonal trends. According to W. Wang et al. (2024), the research showed that in mid-to long-term time series forecasting, GRU outperformed other artificial intelligence methods.

Future maintenance plans will be informed by the wheel replacement, wheel profiling, and MTBF predicting provided by these predictive models.

Based on the top-performing prediction model, a Graphical User Interface (GUI) will be created to guarantee the useful implementation of this research in actual operations. According to (Jansen, 1998), an application that has a good graphical user interface is easy to comprehend and effective to use. By showing the predicted month and year for each action, the GUI will make it simple for maintenance staff to view wheel replacement and profiling forecasts. In order to facilitate more efficient, proactive maintenance planning, this straightforward solution is made to bridge the gap between advanced data analytics and decision-making that takes place on the ground.

Through the integration of deep learning, time-series forecasting, multivariate data analysis, and user-friendly digital tools, this project supports industry-specific requirements as well as national policy objectives. It illustrates how predictive analytics can improve Malaysia's rail infrastructure's long-term sustainability, decrease downtime, and increase dependability.

1.2 Problem Statement

MRT Kajang Line has experienced increased passenger demand, which places greater pressure on train operations and accelerates the rate of wheel wear. Worn-out wheels lead to noticeable passenger discomfort, including increased vibrations and noise due to irregular contact between the wheel and rail surface. Trains with such issues must be withdrawn from service and sent to the depot for wheel profiling or replacement. Hence, it will lead to reduce fleet availability. For example, in situations where it is up to 12 wheels experience issues simultaneously, the number of operational trains becomes insufficient to meet service demands. This disrupts the planned headway, which is the time interval between one train to another train that results in service delays and reduced reliability for commuters.

Currently, forecasting which trains that need wheel profiling or replacement is done manually using Microsoft Excel. While this approach is structured, it is limited in accuracy and efficiency. By developing a predictive model, the process can be automated significantly by enhancing maintenance planning, enhancing forecasting accuracy, and reducing dependency on manual methods. Automation through modelling will also enable faster and data-driven decision making.

As more people use MRT daily, the increased operational load causes the wheels to wear out more quickly. Without a proper system to monitor the wheel condition or predict when failures might occur such as through MTBF prediction, the operations and maintenance teams will face challenges. They will struggle to plan and predict the exact time to order spare parts on time or schedule repairs efficiently, which can result in unexpected train downtimes and higher maintenance costs

1.3 Research Questions

- What are the most significant features for differentiating between failure and non-failure wheels Line using Linear Discriminant Analysis (LDA)?
- How effectively does Holt's Winter Method compare to Gated Recurrent Unit (GRU) in wheel profiling and replacement forecast with Mean Time Before Failure (MTBF) prediction.
- How can a user-friendly graphical user interface (GUI) be designed to show the predicted date for wheel replacement and profiling based on forecasting models?

1.4 Research Objectives

- To identify features that will give best separation failure and non-failure wheels using LDA.
- To forecast and evaluate wheel profiling and replacement using Holt's Winter method and GRU model performance with MTBF prediction.
- To create a graphical user interface (GUI) based on the best-performing model so the employee can easily see the predicted month and year for wheel replacement and profiling.

CHAPTER 2

DATA COLLECTION AND DATA UNDERSTANDING

2.1 Introduction

This chapter will provide an overview of data collection methods and data understanding. This section outlines the data collection process, the source and structure of the dataset, and the selection criteria used to identify relevant train for analysis.

2.2 Data Collection

In this project, the data was obtained from the Rolling Stock Department of MRT Corporation located in MRT Sungai Buloh Depot. The dataset period from 2018 to 2025, with entries recorded every four-month interval to ensure sufficient monitoring and to predict which train needs to be replaced and profiled. The data collection process involved direct collaboration with maintenance analysts through formal discussions and presentations to ensure accurate and relevant information was obtained.

The MRT Kajang Line operates a fleet comprising 58 trains. For this project, a representative sample of 8 trains (T01, T05, T25, T45, T46, T55, T57, and T58) were selected based on maintenance history and availability of complete records. The dataset includes wheel condition indicators such as Wheel Diameter is less than 755 mm and / or Wheel Flange thickness is less than 27.5 mm. These thresholds were defined by the Operational and Maintenance (O&M) Manual.

After filtering the relevant records, the selected data was exported into a new CSV file for analysis. Additional columns were introduced to indicate whether each observation is met by defined failure criteria by using Python. This new feature allows for detailed failure analysis and supports the development of Mean Time Before Failure (MTBF) prediction and calculation.

2.3 Data Understanding

Table 2.1 shows the description of the dataset used for this project

Table 2.1 Data Understanding

Variable	Data Type	Example	Description
Date	Datetime	“2018-06-22”	The date when the measurement or record was taken.
TrainID	String	“T01”	Unique identifier for the train.
Car	String	“M1”, “T1”	Identifier for the specific car (motor/trailer) within the train.
Bogie	String	“BOG#1”	Identifier for the bogie, which is the frame under the train that holds the wheelset.
Axle	String	“X1-LHS”	Identifier for the axle, which is the rod that connects a pair of wheels.
WheelDiameter	Float	“843.22”	Diameter of the wheel measured before any cutting (mm).
WheelFlange	Float	“29.02”	Measurement of the wheel flange height (mm).
FlangeLoss	Float	“2.78”	Amount of flange material lost due to wear (mm)
FlangeTo32mm	Float	“2.20”	How much amount of flange left before the flange reaches 32mm.

CutToPrime	Float	“16.69”	The amount of material to be cut to restore the wheel to prime condition (mm).
WheelDiameterAfterCut	Float	“826.53”	Diameter of the wheel after cutting (mm).
Failure	Integer	“0/1”	Indicates whether a failure occurred (1 = failure, 0 = no failure).
Quarter	String	“2018Q2”	The year and 3-month period the data is from (e.g. 2019Q3 = July–September 2019).
FailuresPerQuarter	Integer	“11”	Total failures happened in the 3-month period.
DaysInQuarter	Integer	“90”	Total number of days in the respective quarter.
MTBF	Float	“147.27”	Quarter number of days between failures.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter discusses the procedures taken to accomplish the research objectives and gives an overview of the methods implemented to carry out the project. The data science lifecycle which involves data collection, preprocessing, model development, and evaluation were the source of the methodological approach, which ensures accurate and reliable wheel profiling and replacement forecasting using Holt-Winters method and GRU model performance with MTBF prediction.

3.2 Project Plan

The project's overall plan and procedure are described in this section. The project progress flow chart which shows the arrangement of tasks involved was displayed in Figure 3.2.

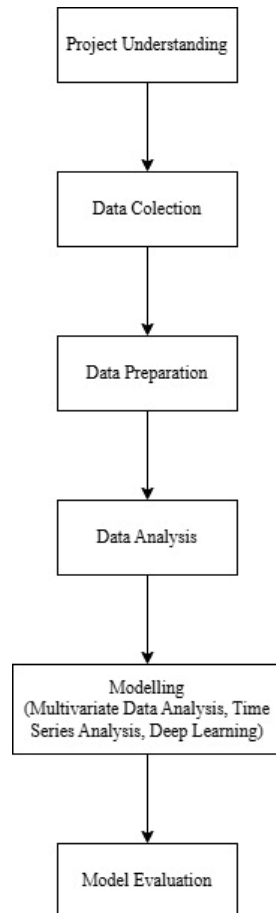


Figure 3.2: Flow Chart of Project Progress.

The structured project progress used in this study is shown in the flow chart in Figure 3.2. Project Understanding was the first step in this process, where the main objectives and scope of the problem were determined. Data collection followed next, during which appropriate datasets were gathered. The data was then cleaned, transformed, and organised in order to make it ready for analysis. This process is known as data preparation. Exploratory methods were used in the Data Analysis phase to find patterns and trends that helped choose the best modelling methods.

Three methods were used in the modelling stage which are Linear Discriminant Analysis (LDA) for multivariate classification, Holt-Winters Method for time series forecasting, and Gated Recurrent Unit (GRU) networks for deep learning-based predictions. Different aspects of the failure data were addressed by each model. Multiple performance metrics, including MAE, MSE, RMSE, MAPE, and R2 score, were used in the final stage, Model Evaluation, to evaluate the models' accuracy and reliability. Effective predictive modelling results and methodical data processing were guaranteed by this thorough project flow.

3.3 Data Preparation

Data preparation ensures the dataset is clean and ready for analysis. Only specific trains and important parameters were included in the project after raw wheel data was filtered and processed. The modifications needed were applied, and missing values were resolved. For accurate failure forecasting and prediction, this stage is crucial.

	Date	TrainID	Car	Bogie	Axle	WheelDiameter	WheelFlange	FlangeLoss	FlangeTo32mm	CutToPrime	WheelDiameterAfterCut
0	22/6/2018	T01	M1	BOG#1	X1-LHS	843.22	29.02	0.00	2.98	16.69	826.53
1	22/6/2018	T01	M1	BOG#1	X1-RHS	843.93	29.26	0.00	2.74	15.34	828.59
2	22/6/2018	T01	M1	BOG#1	X2-LHS	844.14	29.80	0.00	2.20	12.32	831.82
3	22/6/2018	T01	M1	BOG#1	X2-RHS	844.76	29.53	0.00	2.47	13.83	830.93
4	22/6/2018	T01	M1	BOG#2	X3-LHS	844.86	29.80	0.00	2.20	12.32	832.54
...	...	...	...	...	...	...	...	...	...	...	...
5019	13/3/2025	T58	M2	BOG#1	X2-RHS	818.00	30.07	0.56	1.93	10.80	807.20
5020	13/3/2025	T58	M2	BOG#2	X3-LHS	816.30	30.52	0.23	1.48	8.27	808.03
5021	13/3/2025	T58	M2	BOG#2	X3-RHS	812.10	30.00	2.78	2.00	11.21	800.89
5022	13/3/2025	T58	M2	BOG#2	X4-LHS	812.10	30.01	-1.17	1.99	11.14	800.96
5023	13/3/2025	T58	M2	BOG#2	X4-RHS	808.10	30.03	1.59	1.97	11.04	797.06

5024 rows × 11 columns

Figure 3.4.1: Table of the Wheel Dataset.

The structured table of the wheel dataset applied for analysis in this study is shown in Figure 3.4.1. Date, TrainID, Car, Bogie, Axle, WheelDiameter, WheelFlange, FlangeLoss, FlangeTo32mm, CutToPrime, and WheelDiameterAfterCut are among its key features. It has 5024 rows and 11 columns.

df.dtypes	
	0
Date	object
TrainID	object
Car	object
Bogie	object
Axle	object
WheelDiameter	float64
WheelFlange	float64
FlangeLoss	float64
FlangeTo32mm	float64
CutToPrime	float64
WheelDiameterAfterCut	float64
dtype: object	

Figure 3.4.2: Data Types before change.

df.dtypes	
	0
Date	datetime64[ns]
TrainID	object
Car	object
Bogie	object
Axle	object
WheelDiameter	float64
WheelFlange	float64
FlangeLoss	float64
FlangeTo32mm	float64
CutToPrime	float64
WheelDiameterAfterCut	float64
dtype: object	

Figure 3.4.3: Data Types after change.

```
df['Date'] = pd.to_datetime(df['Date'], format='%d/%m/%Y')
```

Figure 3.4.4: Change format of Date column into datetime format

The dataset's data types are displayed in Figure 3.4.2 before any modifications. Figure 3.4.4 displays the code used to convert the Date column into datetime format. After the Date column was successfully updated, the new data types are shown in Figure 3.4.3. In order to enable appropriate time-based analysis, sorting, and filtering, the Date column must be converted into datetime format. Furthermore, it allows to use for time-series forecasting models like Holt-Winters Method and GRU that depend on temporal trends.

```
# Drop the column 'FlangeLoss, FlangeTo32mm, CutToPrime, WheelDiameterAfterCut'
df = df.drop(columns=['FlangeLoss', 'FlangeTo32mm', 'CutToPrime', 'WheelDiameterAfterCut'])
```

Figure 3.4.5: Drop unrelated columns.

df							
	Date	TrainID	Car	Bogie	Axle	WheelDiameter	WheelFlange
0	2018-06-22	T01	M1	BOG#1	X1-LHS	843.22	29.02
1	2018-06-22	T01	M1	BOG#1	X1-RHS	843.93	29.26
2	2018-06-22	T01	M1	BOG#1	X2-LHS	844.14	29.80
3	2018-06-22	T01	M1	BOG#1	X2-RHS	844.76	29.53
4	2018-06-22	T01	M1	BOG#2	X3-LHS	844.86	29.80
...	...	...	...	...	...	...	...
5019	2025-03-13	T58	M2	BOG#1	X2-RHS	818.00	30.07
5020	2025-03-13	T58	M2	BOG#2	X3-LHS	816.30	30.52
5021	2025-03-13	T58	M2	BOG#2	X3-RHS	812.10	30.00
5022	2025-03-13	T58	M2	BOG#2	X4-LHS	812.10	30.01
5023	2025-03-13	T58	M2	BOG#2	X4-RHS	808.10	30.03
5024 rows × 7 columns							

Figure 3.4.6: Dataset after remove the unrelated column.

The code used to remove irrelevant columns from the dataset since they are not related to the project's objectives is displayed in Figure 3.4.5. The dataset becomes simplified and more effective for analysis and modelling when unnecessary data is removed. After the successful removal of the irrelevant columns, the updated dataset is shown in Figure 3.4.6. This ensures that only the most important features remain for accurate forecasting and prediction.

```
#check duplicate data
df.duplicated().sum()

np.int64(0)
```

Figure 3.4.7: Duplicate Data.

```
#check missing value
df.isnull().sum()

0
Date      0
TrainID    0
Car        0
Bogie      0
Axle       0
WheelDiameter  0
WheelFlange  0

dtype: int64
```

Figure 3.4.8: Missing Value Data.

Figure 3.4.7 shows that there are no duplicate entries in the dataset. Moreover, Figure 3.4.8 indicates that every column has no missing values.

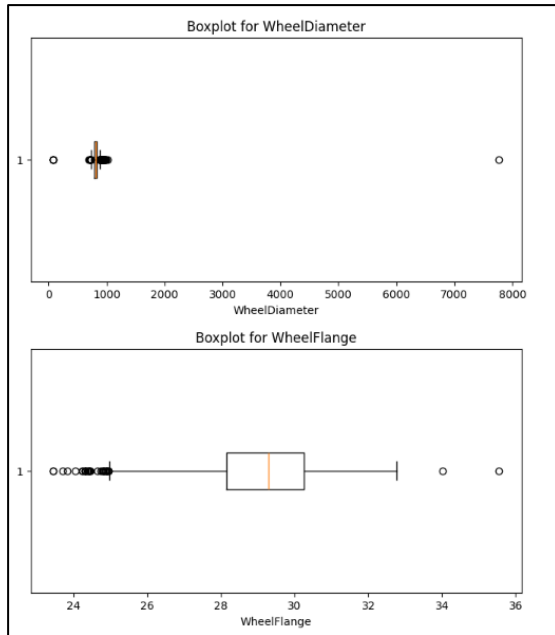


Figure 3.4.9: Boxplot before Cleaning.

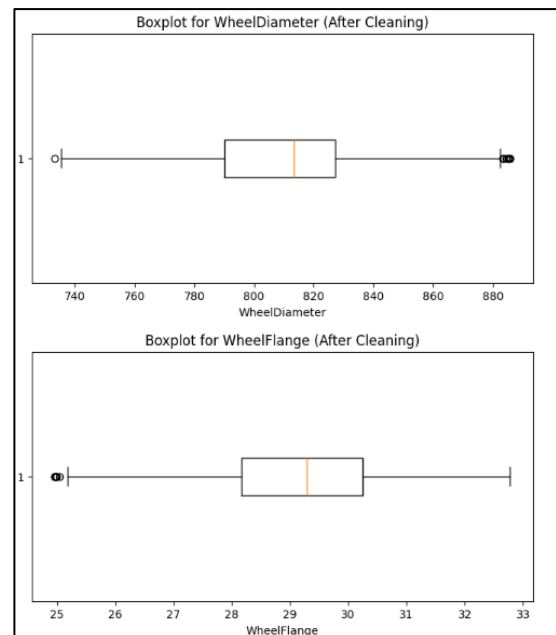


Figure 3.4.10: Boxplot after Cleaning.

Outliers in the dataset were identified using a boxplot, as shown in Figure 3.4.9, which displays the data before cleaning. To maintain data integrity, these outliers were identified and removed using the Interquartile Range (IQR) method. After the outliers were eliminated, the boxplot which displays a more uniform distribution was shown in Figure 3.4.10. By reducing the impact of extreme values, this step improves in increasing the forecasting models' accuracy.

```
# Check total number of attributes
print("No. of Attributes (Columns): ",len(df.columns))
print("No. of sample (Rows): ",len(df))

No. of Attributes (Columns): 7
No. of sample (Rows): 4923
```

Figure 3.4.11: Row and columns after remove outliers.

Figure 3.4.11 shows the number of columns and rows in the dataset after outlier removal. The dataset now contains 7 columns and 4,923 rows.

```
# Failure occurs if WheelDiameter < 755 OR WheelFlange < 27.5 (covers all failure scenarios)
df['Failure'] = df.apply(
    lambda row: 1 if (row['WheelDiameter'] < 755) or (row['WheelFlange'] < 27.5) else 0,
    axis=1
)
df
```

Figure 3.4.12: Column of failure.


```
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
df['Quarter'] = df['Date'].dt.to_period('Q').astype(str)
```

Figure 3.4.13: Column of Quarter.

```
# Calculate failures per quarter
failures_per_quarter = df.groupby('Quarter')['Failure'].sum().reset_index(name='FailuresPerQuarter')

# Merge back to original DataFrame, keeping only 'Quarter' and 'FailuresPerQuarter'
df = df.merge(
    failures_per_quarter[['Quarter', 'FailuresPerQuarter']], # Select only these columns
    on='Quarter', # Merge key
    how='left' # Preserve all original rows
)

# Show the updated DataFrame
df
```

Figure 3.4.14: Column of Failure per Quarter.

```
# Function to calculate days in a quarter
def get_days_in_quarter(q_str):
    q = pd.Period(q_str)
    return (q.end_time - q.start_time).days + 1

df['DaysInQuarter'] = df['Quarter'].apply(get_days_in_quarter)
df
```

Figure 3.4.15: Column of Days in Quarter.

```
# Corrected MTBF calculation (use 'FailuresPerQuarter' instead of 'Failure')
df['MTBF'] = (18 * df['DaysInQuarter']) / df['FailuresPerQuarter']
df['MTBF'] = df['MTBF'].replace([np.inf, -np.inf], np.nan)
df
```

Figure 3.4.16: Column of MTBF.

```
df['MTBF'] = df['MTBF'].fillna(0)
df
```

Figure 3.4.17: Replace NaN with 0 in MTBF column.

	Date	TrainID	Car	Bogie	Axle	WheelDiameter	WheelFlange	Failure	Quarter	FailuresPerQuarter	DaysInQuarter	MTBF
0	2018-06-22	T01	M1	BOG#1	X1-LHS	843.22	29.02	0	2018Q2	0	91	0.000000
1	2018-06-22	T01	M1	BOG#1	X1-RHS	843.93	29.26	0	2018Q2	0	91	0.000000
2	2018-06-22	T01	M1	BOG#1	X2-LHS	844.14	29.80	0	2018Q2	0	91	0.000000
3	2018-06-22	T01	M1	BOG#1	X2-RHS	844.76	29.53	0	2018Q2	0	91	0.000000
4	2018-06-22	T01	M1	BOG#2	X3-LHS	844.86	29.80	0	2018Q2	0	91	0.000000
...	...	...	...	...	...	...	...	...	...	...	...	...
4918	2025-03-13	T58	M2	BOG#1	X2-RHS	818.00	30.07	0	2025Q1	11	90	147.272727
4919	2025-03-13	T58	M2	BOG#2	X3-LHS	816.30	30.52	0	2025Q1	11	90	147.272727
4920	2025-03-13	T58	M2	BOG#2	X3-RHS	812.10	30.00	0	2025Q1	11	90	147.272727
4921	2025-03-13	T58	M2	BOG#2	X4-LHS	812.10	30.01	0	2025Q1	11	90	147.272727
4922	2025-03-13	T58	M2	BOG#2	X4-RHS	808.10	30.03	0	2025Q1	11	90	147.272727

4923 rows x 12 columns

Figure 3.4.18: Dataset after all the column added.

As shown in Figure 3.4.12, the Failure column was added. If the wheel diameter is less than 755 mm, the wheel flange thickness is less than 27.5 mm or both are met, then a failure is recorded. As demonstrated in Figure 3.4.13, a Quarter column was developed based on the date in order to arrange the data sequentially. The total number of failures in each quarter was then determined by adding a Failures Per Quarter column, as shown in Figure 3.4.14. As shown in Figure 3.4.15, a Days In Quarter column reflecting the number of days in each quarter was developed to support in the MTBF calculations. Lastly, as seen in Figure 3.4.16, the formula $(18 \times \text{Days in Quarter}) / \text{Failures Per Quarter}$ was used to determine the MTBF (Mean Time Between Failures) column. Any resulting Not a Number (NaN) values were substituted with 0 as shown in Figure 3.4.17. After the successful addition of all new columns, the final dataset is shown in Figure 3.4.18.

```
# Save the updated dataset to a new CSV file
output_file_path = r"C:\Users\ASUS\OneDrive - ump.edu.my\Documents\UMP SEM 6\GROUP PROJECT SEM 6"
df.to_csv(output_file_path, index=False)
print(f"Cleanded Data saved to: {output_file_path}")
```

Figure 3.4.19: Save csv file.

Figure 3.4.19 shows the process of saving the updated dataset into a CSV file to ensure the cleaned and processed data, including all newly added columns, can be reused or shared for further analysis and modelling.

3.4 Exploratory Data Analysis

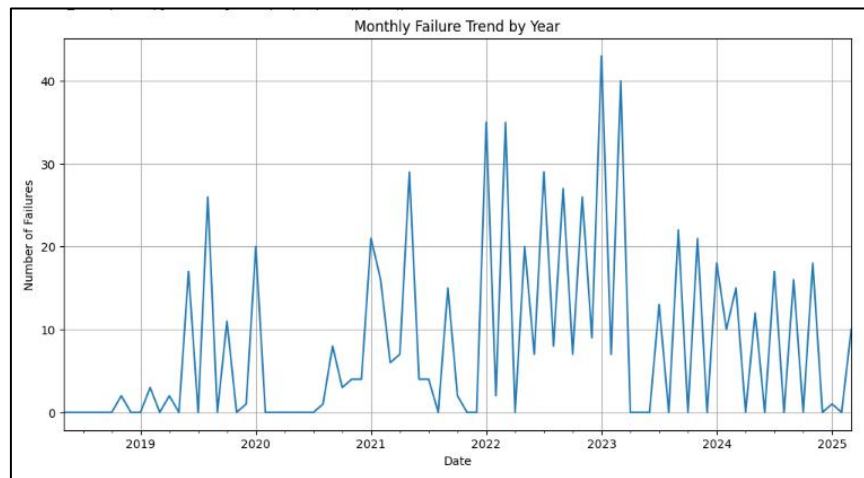


Figure 3.4.1: Line chart of monthly failure trend by year.

This line chart of the Monthly Failure Trend by Year illustrates the number of wheel failures recorded each month from 2018 to early 2025 as shown in the Figure 3.4.1. According to the trend, failures started increased around 2020 and the first few months of 2023, with some months recorded over 40 failures. After that, the trend indicates a decrease in the frequency of failures, while there are still a few differences in 2024. There have been relatively few failures in the recent few months of 2025, which may be the result of better maintenance, changes to operations, or insufficient data for the year. Overall, the chart indicates that failure rates differed significantly over time, with obvious peaks indicating times that may require further investigation.

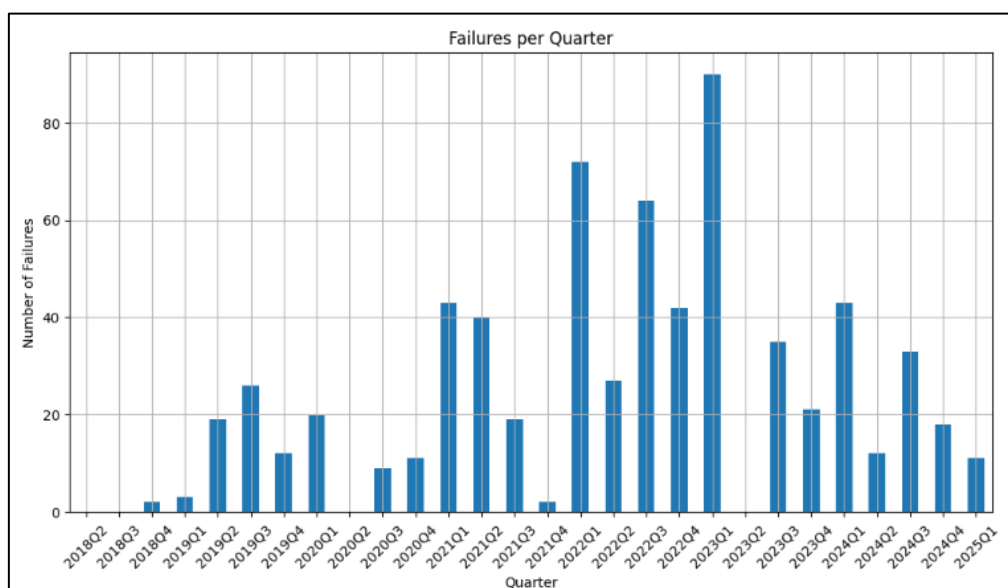


Figure 3.4.2: Bar chart of failures per quarter.

This bar chart illustrates the failures per quarter from Q2 2018 to Q1 2025 as shown in the Figure 3.4.2. Between 2018 and 2021, there was an upward trend in the number of failures, with significant increases in Q2 2021 and Q1 2022, and the greatest number of failures ever recorded in Q1 2023. Although it fluctuates over the next quarters, the number of failures shows a decreasing trend after that peak. This pattern may indicate enhanced operating or maintenance strategies being in place after the beginning of 2023.

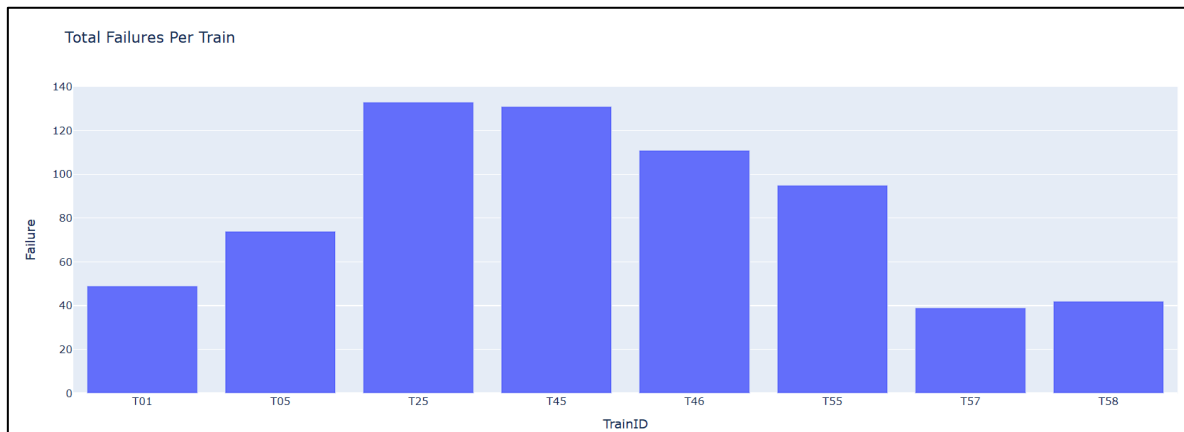


Figure 3.4.3: Total Failures Per Train

The bar chart in Figure 3.4.3 illustrates the number of recorded failures for each train, identified by their respective TrainID labels. Among the trains shown, Train T25 and Train T45 shows the highest number of failures, each recording totals exceeding 130 failures, indicating a significantly higher rate failure issues in wheels. Train T46 and T55 follow, with slightly lower but still notable failure counts that range from approximately 95 to 110 failures. In contrast, trains such as T01, T57, and T58 have experienced comparatively fewer failures, all reporting below 50 failures. These findings suggest that certain trains may require closer inspection, more frequent maintenance, or further investigation into recurring issues. Identifying trains with the highest failure counts can help prioritize maintenance efforts and improve overall fleet reliability.

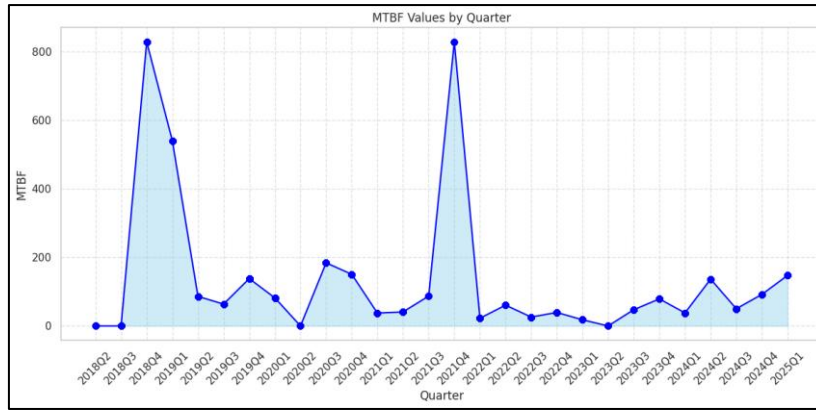


Figure 3.4.4: MTBF values by quarter.

The Mean Time Before Failure (MTBF) figures each quarter from 2018 to 2025 are shown in this Figure 3.4.4, which indicates significant differences in wheel performance over time. Q4 2018 and Q4 2021 showed significant MTBF the highest points, suggesting prolonged wheel durability, possibly due to reduced operations or effective maintenance. In contrast, there were significant decreases in a number of quarters, including Q2 2020 and Q1 2023, which indicated an increased in wheel failures. After 2022, although the fact that MTBF values remained quite low. This suggests that maintenance procedures or operational variables have recently improved. Overall, this analysis shows how wheel lifespan differs and supports the need for accurate forecasting models to schedule replacement and profiling in the correct time frame.

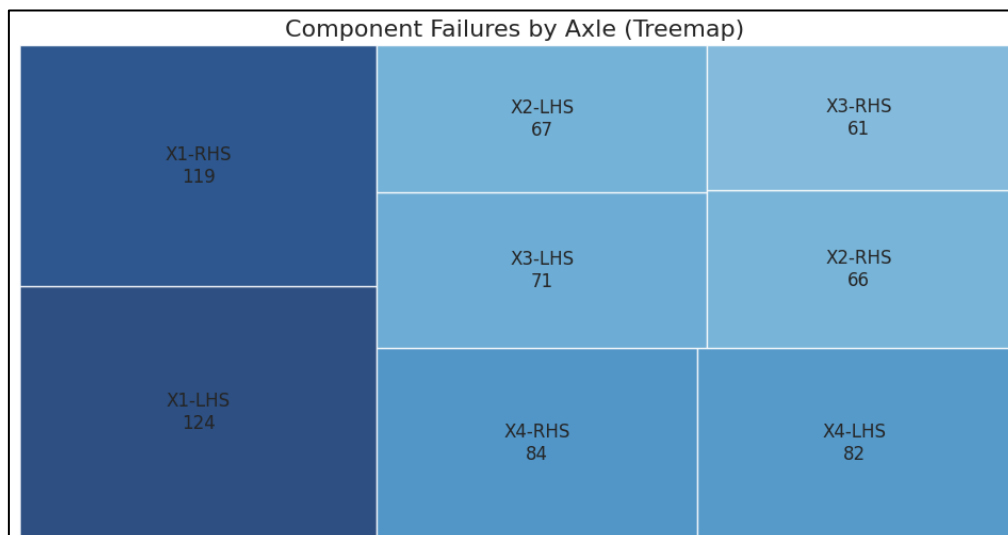


Figure 3.4.5: Tree map of component failures by axle.

According to Figure 3.4.5 show the tree map of component failures by axle, Axle X1 has a high concentration of component failures. Its Left-Hand Side (LHS) has 124 failures and Right-Hand Side (RHS) has 119 failures both significantly contribute to this failure, making it the most common failure position which 36% of the 674 total failures. Significant failure rates with LHS are 82 and RHS are 84 are also recorded by Axle X4, suggesting another crucial area. Front axles X1 and X2 with 376 failures are significantly more likely to fail than rear axles X3 and X4 with 298 failures, which may indicate that the front is more pressured or more sensitive. Axle X3 has a significant left-side imbalance with LHS is 71 between RHS is 61, indicating possible asymmetric wear or external factors, but failures on X2 and X4 have a comparable distribution between the sides. Priority targets for analysing the root cause are identified as X1 of both sides, X4 RHS, and X3 LHS, indicating systemic or operational issues that significantly impact front axles along with certain sides.

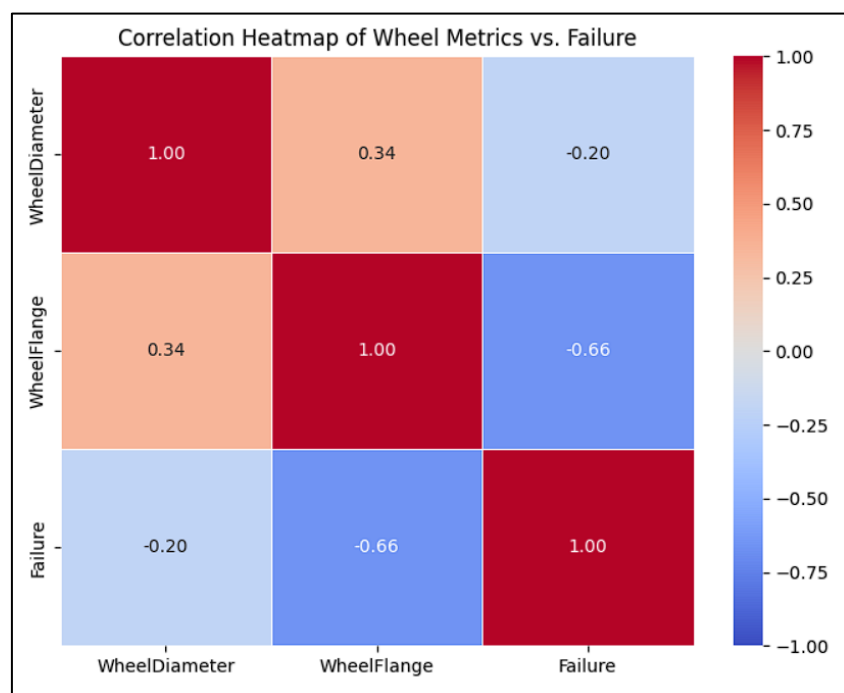


Figure 3.4.6: Correlation Heatmap of Wheel Diameter, Wheel Flange and Failure

The correlation heatmap in Figure 3.4.6 provides insight into the relationships between Wheel Diameter and Wheel Flange also occurrence of failures. The correlation coefficient between Wheel Flange and Failure is -0.66, indicating a strong negative correlation. This suggests that as the wheel flange height decreases, the failure increases significantly. In contrast, the correlation between Wheel Diameter and Failure is -0.20, reflecting a weak

negative relationship, meaning that smaller wheel diameters are only slightly associated with higher failure rates. Additionally, there is a moderate positive correlation (0.34) between Wheel Diameter and Wheel Flange, implying that larger wheels tend to have slightly higher flanges. Overall, these findings highlight wheel flange wear as a critical factor in predicting failures, emphasizing the importance of regular inspection and maintenance to mitigate operational risks.

3.5 Modelling

Three techniques were employed in this study to achieve the project objectives which are Linear Discriminant Analysis (LDA) was utilized to identify the most effective features for giving best separation between failure and non-failure wheels, Holt's Method and Long Short-Term Memory (LSTM) models were applied to forecast and evaluate wheel profiling and replacement with MTBF prediction. These techniques provide a comprehensive approach to classification and forecasting to ensure accurate prediction and efficient decision-making for wheel maintenance and operation planning

3.5.1 Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) is a classification technique used to find the combination of features that best separates two or more predefined groups. In this study, LDA was used to identify which wheel condition features most effectively separates between failure and non-failure cases. This step was essential to better understand the underlying patterns in the data and to guide the development of more accurate predictive models.

The analysis focused on key wheel measurements and revealed clear differences in average values between failed and non-failed wheels. LDA computed a linear function that maximizes the separation between these two groups. Among the features considered, some contributed more strongly to the separation than others, indicating their higher importance in predicting potential failures.

By projecting the data onto a new axis based on the linear discriminant, LDA not only improved class separability but also reduced the complexity of the dataset. This enhanced the overall efficiency of the modelling process and provided clearer insights into which factors

should be prioritized when monitoring wheel health. The results from LDA served as a foundation for subsequent forecasting and decision-support tools developed in this study.

3.5.2 Holt-Winters method

The Holt-Winters Method, sometimes referred to as Triple Exponential Smoothing, is an extension of Holt's Method that is applied when time series data shows both seasonality and a trend. By implementing a third equation to handle for the seasonal component, this model improves forecasting accuracy and is suitable for datasets like wheel diameter and flange thickness that may show cyclical wear patterns as a result of maintenance cycles or operational schedules.

The Holt-Winters Method is used in this project to predict when wheel diameter and flange thickness would likely decrease below failure limits, requiring replacement or profiling. Since the data shows clear seasonal trend across time, this method works more effectively than the straightforward Holt's Method. The model incorporates three key components which is level that representing the baseline value at a given time, trend that indicating the direction and rate of change and seasonality that capturing recurring short-term patterns. This enables it to model and forecast wheel measurement wear progression with accuracy. These components are updated at each time step using three smoothing parameters which are gamma (γ), beta (β), and alpha (α). This allows the model to adapt to seasonal changes in wear rate dynamically.

Historical and previous wheel data (2018-2025) from certain MRT Kajang Line trains is used to apply Holt-Winters Method. Wheel diameter and flange thickness time-stamped measurements gathered over several maintenance intervals are included in this. The model is trained on this data to detect patterns of wear progression and then used to project future values. To determine the amount of time before failure, forecasts are then compared to the defined failure thresholds, which are 27.5 mm for flange thickness and 775 mm for wheel diameter.

Based on historical measurements of wheel diameter and flange thickness, Holt-Winters Method used to our project to forecast when wheel replacement or profiling will be necessary. This helps in the maintenance team's preparation and immediate response to avoid malfunctions. Better scheduling of maintenance tasks is made possible by Holt-Winters Method, which analyses historical wear patterns and trends over time to provide insights into future wheel conditions. In addition to reducing maintenance budgets and minimising unplanned downtime, this guarantees the safety, effectiveness, and dependability of train operations.

3.5.3 Gated Recurrent Unit (GRU)

A Gated Recurrent Unit (GRU) is a type of Recurrent Neural Network (RNN) designed to effectively model sequential data while being computationally simpler than Long Short-Term Memory (LSTM) networks. GRU overcomes challenges like the vanishing gradient problem by using gating mechanisms especially the update and reset gates which help regulate information flow and preserve important patterns over time. This makes GRU well-suited for time-series forecasting, especially when long-term memory is needed but model efficiency is also a concern.

This project uses GRU to predict wheel diameter and flange thickness over time using measurement and maintenance data from specific MRT Kajang Line trains. GRU offers an effective method for forecasting when wheel measurements are going to drop below the safety thresholds, suggesting the need for replacement or profiling, because it effectively learns from complicated temporal patterns and non-linear degradation trends in the data. In comparison to traditional linear models, GRU can better handle fluctuations, anomalies, and varying wear rates in wheel data.

The GRU model is trained using preprocessed wheel diameter and flange thickness time-series sequences. To assess accuracy, the dataset is divided into training and testing sets. Normalization is then used to enhance model performance. To maximise outcomes, hyperparameters such the number of GRU units, layers, learning rate, and batch size are adjusted. The success of the model is then evaluated by compared its predictions with actual measurements.

Based on historical data, the GRU model may be implemented in our project to precisely predict future wheel conditions, assisting the maintenance team in taking preventive steps before failures happen. The model offers insights into when wheel replacement or profiling should be planned by identifying patterns in historical wheel wear trends. This makes it possible to better allocate maintenance supplies, reduces unexpected breakdowns, and ensures the long-term safety and effectiveness of train operations. The company can prolong the service life of wheel components and optimise its maintenance strategy with more accurate forecasts.

3.6 Evaluation

To assess the performance and accuracy of the forecasting models, several standard error metrics were used. These metrics provide quantitative measures of how closely the model's predictions align with the actual observed values. The following evaluation criteria were employed:

1. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where:

n = total number of observations

y_i = actual value at instance i

$\hat{y}_i$ = predicted value at instance i

MAE measures the average magnitude of the errors in a set of predictions, without considering their direction. It is a linear score which means all the individual differences are weighted equally.

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE measures the average of the squared differences between actual and predicted values. It penalizes larger errors more severely than MAE.

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE is the square root of the MSE. It retains the same unit as the original data and provides a measure of the typical size of the errors.

4. Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MAPE expresses the prediction error as a percentage, offering an intuitive understanding of error relative to the actual values. It is especially useful when comparing forecasting accuracy across different datasets.

5. Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

R^2 indicates how well the model explains the variability of the target variable. A score of 1.0 indicates perfect prediction, while a score of 0 indicates that the model does not explain any variability in the data.

These metrics were used collectively to evaluate and compare the performance of the Holt Winters Method model and the GRU model. The model with lower error values (MAE, MSE, RMSE, MAPE) and a higher R^2 score was considered more reliable for forecasting wheel profiling and replacement.

3.7 Conclusion

In conclusion, this chapter provided the overall method required to complete the project, starting with understanding the project's scope and moving on to systematic steps including data collection, preparation, and analysis. Models based on the processed data have been developed and evaluated using an organised methodology. Every stage of the process was created to guarantee the accuracy and dependability of the results. This structured flow provides the basis for the analysis and findings in the following chapter by offering an organised framework to guide the entire project process.

CHAPTER 4

DATA ANALYSIS, RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the analysis and results based on the project objectives. Each section discusses the method used, findings obtained, and how the results align with the aim of improving the railway operations.

4.2 Multivariate Data Analysis

```
Call:
lda(Failure ~ WheelDiameter + WheelFlange, data = data)

Prior probabilities of groups:
      0      1
0.8630916 0.1369084

Group means:
  WheelDiameter WheelFlange
0      812.9757    29.60291
1      798.7814    26.73681

Coefficients of linear discriminants:
              LD1
WheelDiameter  0.002629184
WheelFlange   -0.898452568
```

Figure 4.2.1: Develop LDA model

Based on model summary on Figure 4.2.1, the LDA model was used to classify the Failure variable based on two predictors: Wheel Diameter and Wheel Flange. The prior probabilities indicate that 86.31% of the observations belong to group 0 (non-failure), while 13.69% belong to group 1 (failure), showing a class imbalance. The group means show that group 0 has higher average values for both Wheel Diameter (812.98) and Wheel Flange (29.60) compared to group 1 (798.78 and 26.74, respectively). The coefficients of the linear discriminant function indicate the importance of each variable in separating the groups. Since there are only two groups, the model produces one discriminant function (LD1). The equation for LD1 is:

$$LD1 = 0.002629 \times WheelDiameter - 0.898453 \times WheelFlange$$

This formula above is used to calculate a score for each observation, which helps determine its classification. A higher Wheel Diameter slightly increases the LD1 score, while a higher Wheel Flange strongly decreases it. The large negative coefficient for Wheel Flange suggests that it is the most influential variable in distinguishing between failure and non-failure cases. The model uses these scores to assign each observation to the group with the closest mean value along the LD1 axis.

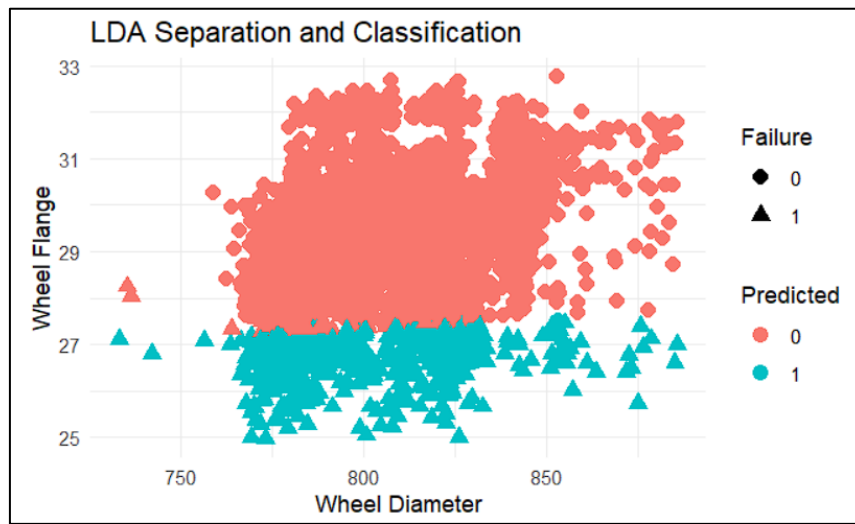


Figure 4.2.2: LDA Separation and Classification Output

The scatter plot visually displays in Figure 4.2.2 which is the separation between the two failure groups based on the predictor variables. Each point represents an observation, with actual group labels shown using black diamonds (group 0) and black triangles (group 1), while predicted classifications are shown using red circles (predicted group 0) and cyan triangles (predicted group 1). The plot shows that group 0 observations are mostly located in the upper region with higher Wheel Flange values, while group 1 observations tend to cluster in the lower region with smaller Wheel Flange values. This visual pattern aligns with the LDA output, where Wheel Flange was identified as the stronger predictor. The model generally performs well in separating the two groups, although some overlap and misclassifications are observed. These visual results support the conclusion that Wheel Flange is a key variable in distinguishing failure status and that the LDA model is effective despite some classification errors.

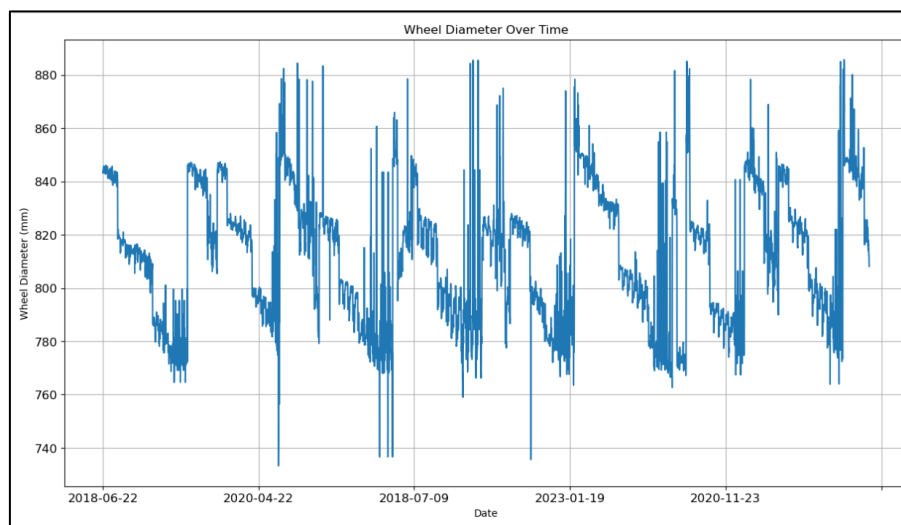
4.3 Time Series Analysis

The objective of this project is to determine three important railway maintenance metrics such as wheel diameter, wheel flange thickness, and Mean Time Between Failures (MTBF) using the Holt-Winters (Triple Exponential Smoothing) method. For trains to operate safely and effectively. Maintenance teams can plan early interventions, avoid unplanned failures, and allocate resources more efficiently when these values are accurately forecasted. This study uses historical time series data for simulating the seasonality and trend existent in each variable, allowing proactive maintenance planning and better decision-making.

Furthermore, to make sure the data could be utilised efficiently for time series analysis, the date column was first transformed into a datetime format. A training set and a testing set were then created from the entire dataset. The training set consists of the first 4,429 observations, which serve as the in-sample data for model fitting as shown in the Figure 4.3.1. To evaluate the predicted ability of the model, any extra data outside of this range is referred to as out-of-sample data for testing data. To visually examine trends, seasonality, and any abnormalities in the time series, plots were created for every variable. These plots revealed noticeable patterns, including cyclical changes that suggested the presence of quarterly seasonality, particularly in wheel wear and MTBF variations as shown in the Figure 4.3.2.

```
training = df[0:4429] # in-sample data (training data)
testing = df[4429:] #out-sample data (testing data)
```

Figure 4.3. 1 Data Splitting.



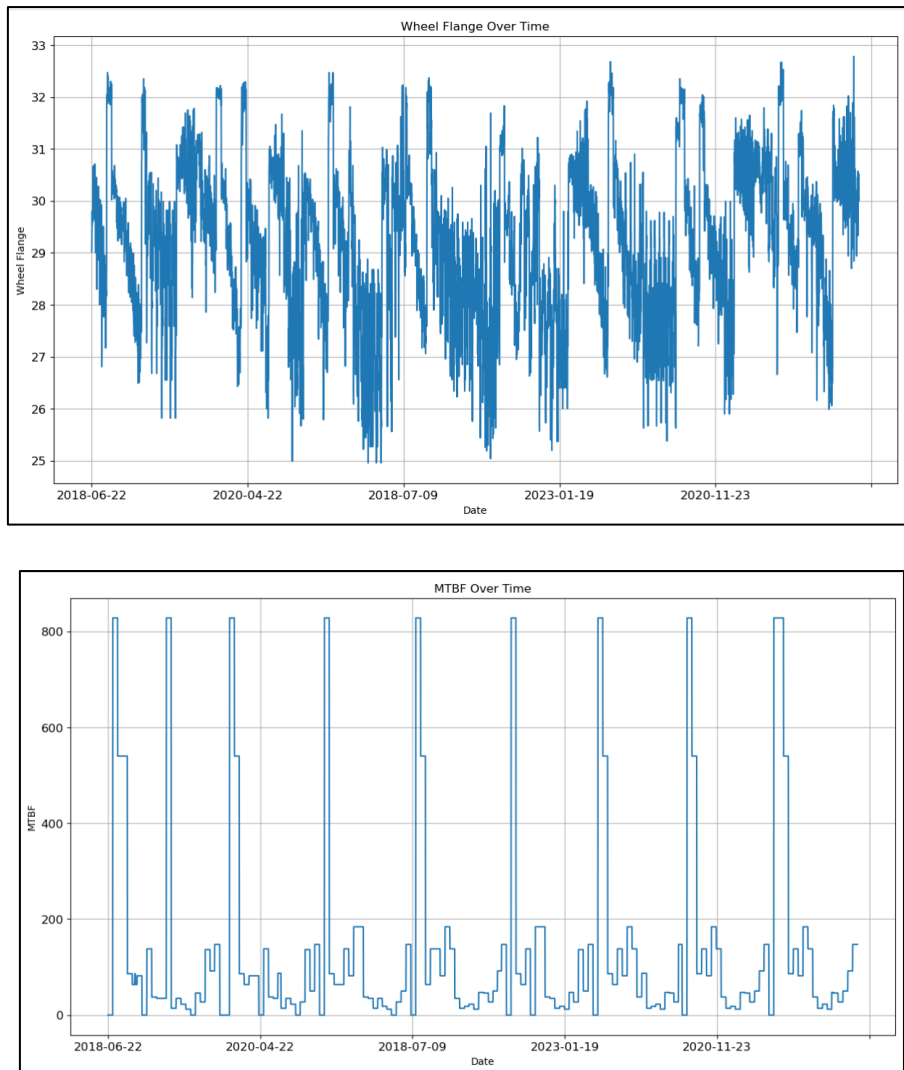


Figure 4.3. 2 Plot of time series data for wheel diameter, wheel flange and MTBF.

The Holt-Winters Exponential Smoothing technique was applied using the Python statsmodels library in order to efficiently capture the trend and seasonal components of the data. This method works well with time series that show long-term trends and seasonal behaviour. An additive model was selected for all three variables based on visual inspection of the data and the nature of the observed patterns, which did not indicate multiplicative seasonality. The seasonal period was set to four, assuming quarterly changes are representative of periodic fluctuations in wheel and MTBF data as shown in the Figure 4.3.3.

```
from statsmodels.tsa.api import ExponentialSmoothing, SimpleExpSmoothing, Holt

# Fit triple exponential smoothing model
model1 = ExponentialSmoothing(training_wd['WheelDiameter'], trend='add', seasonal='add', seasonal_periods=4).fit()
model1.summary()
```



```
# Fit triple exponential smoothing model
model3 = ExponentialSmoothing(training_wf['WheelFlange'], trend='add', seasonal='add', seasonal_periods=4).fit()
model3.summary()
```

```
# Fit triple exponential smoothing model
model2 = ExponentialSmoothing(training_wd['MTBF'], trend='add', seasonal='add', seasonal_periods=4).fit()
model2.summary()
```

Figure 4.3. 3 Library of statmodels used and seasonal period was set to four.

ExponentialSmoothing Model Results			
Dep. Variable:	WheelDiameter	No. Observations:	4429
Model:	ExponentialSmoothing	SSE	474023.974
Optimized:	True	AIC	20713.089
Trend:	Additive	BIC	20764.257
Seasonal:	Additive	AICC	20713.139
Seasonal Periods:	4	Date:	Tue, 10 Jun 2025
Box-Cox:	False	Time:	14:10:16
Box-Cox Coeff.:	None		
	coeff	code	optimized
smoothing_level	0.3959666	alpha	True
smoothing_trend	0.0002235	beta	True
smoothing_seasonal	0.0001538	gamma	True
initial_level	844.50750	l.0	True
initial_trend	0.0112509	b.0	True
initial_seasons.0	0.0761035	s.0	True
initial_seasons.1	-0.1566933	s.1	True
initial_seasons.2	0.0305500	s.2	True
initial_seasons.3	0.0418483	s.3	True

ExponentialSmoothing Model Results			
Dep. Variable:	WheelFlange	No. Observations:	4429
Model:	ExponentialSmoothing	SSE	2703.732
Optimized:	True	AIC	-2169.892
Trend:	Additive	BIC	-2118.724
Seasonal:	Additive	AICC	-2169.842
Seasonal Periods:	4	Date:	Tue, 10 Jun 2025
Box-Cox:	False	Time:	14:19:44
Box-Cox Coeff.:	None		
	coeff	code	optimized
smoothing_level	0.5587588	alpha	True
smoothing_trend	6.1235e-19	beta	True
smoothing_seasonal	0.0314599	gamma	True
initial_level	29.343908	l.0	True
initial_trend	0.0006360	b.0	True
initial_seasons.0	-0.2193989	s.0	True
initial_seasons.1	-0.0819115	s.1	True
initial_seasons.2	0.0741031	s.2	True
initial_seasons.3	0.1033201	s.3	True

ExponentialSmoothing Model Results			
Dep. Variable:	MTBF	No. Observations:	4923
Model:	ExponentialSmoothing	SSE	10249337.468
Optimized:	True	AIC	37632.890
Trend:	Additive	BIC	37684.904
Seasonal:	Additive	AICC	37632.935
Seasonal Periods:	4	Date:	Tue, 10 Jun 2025
Box-Cox:	False	Time:	14:14:59
Box-Cox Coeff.:	None		
	coeff	code	optimized
smoothing_level	0.9950041	alpha	True
smoothing_trend	9.9918e-05	beta	True
smoothing_seasonal	9.9835e-05	gamma	True
initial_level	-0.0001770	l.0	True
initial_trend	0.0313509	b.0	True
initial_seasons.0	0.1000640	s.0	True
initial_seasons.1	-0.0308837	s.1	True
initial_seasons.2	-0.0143106	s.2	True
initial_seasons.3	-0.0554399	s.3	True

Figure 4.3.4 Exponential Smoothing Model Results of all target variables.

The exponential smoothing result for all three target variables were shown in the Figure 4.3.4. Regarding wheel diameter series, the model was fitted using parameters that included a smoothing seasonal component at 0.0002, a smoothing trend at 0.0002, and a smoothing level of 0.3995. The model yielded an Akaike Information Criterion (AIC) of 20,712.779 and a Sum of Squared Errors (SSE) of 473,990.73. The smoothing level for the wheel flange was 0.5588, the seasonal component was 0.0315, and the smoothing trend was close to zero. This variable's model produced an AIC of -2,169.892 and an SSE of 2,703.732. In the case of MTBF, which showed higher dynamic fluctuations, the model was highly responsive with a smoothing level of 0.9970, a smoothing trend of 5.9579e-05, and a smoothing seasonal value of 3.5352e-05. The AIC for MTBF was determined to be 37,633.662, whereas the SSE was 10,250,944.899. These results suggest that the Holt-Winters model, which demonstrated very high responsiveness in modelling MTBF, successfully captured the trends in all three measures.

```

# initialize variables for storing predictions and actual values
predictions1 = []
actuals = []

# iterate over each time step in the testing data
for i in range(len(testing_wd)):
    # make one-step ahead forecast
    yhat = model11.forecast()

    # store prediction and actual value
    predictions1.append(yhat)
    actuals.append(testing_wd.iloc[i])

# add actual value to training data
training_wd = pd.concat([training_wd, pd.DataFrame([testing_wd.iloc[i]], columns=training_wd.columns)], ignore_index=True)

```

```

# initialize variables for storing predictions and actual values
predictions3 = []
actuals = []

# iterate over each time step in the testing data
for i in range(len(testing_wf)):
    # make one-step ahead forecast
    yhat = model3.forecast()

    # store prediction and actual value
    predictions3.append(yhat)
    actuals.append(testing_wf.iloc[i])

# add actual value to training data
training_wf = pd.concat([training_wf, pd.DataFrame([testing_wf.iloc[i]], columns=training_wf.columns)], ignore_index=True)

```

```

# initialize variables for storing predictions and actual values
predictions2 = []
actuals = []

# iterate over each time step in the testing data
for i in range(len(testing_wd)):
    # make one-step ahead forecast
    yhat = model12.forecast()

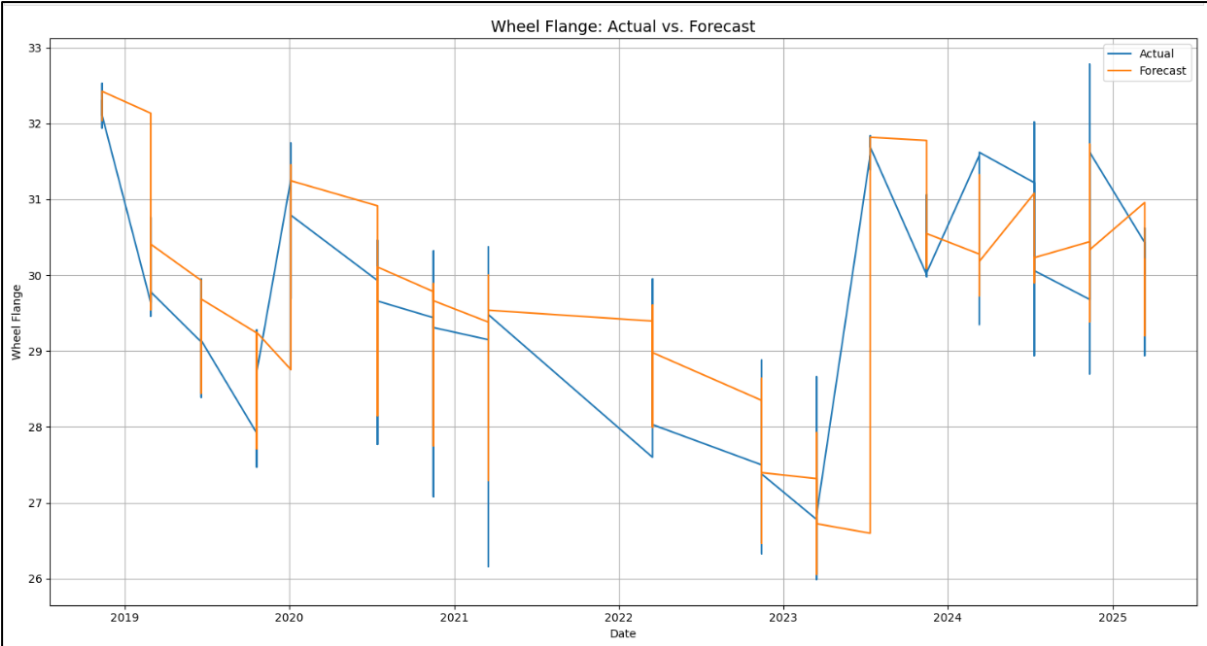
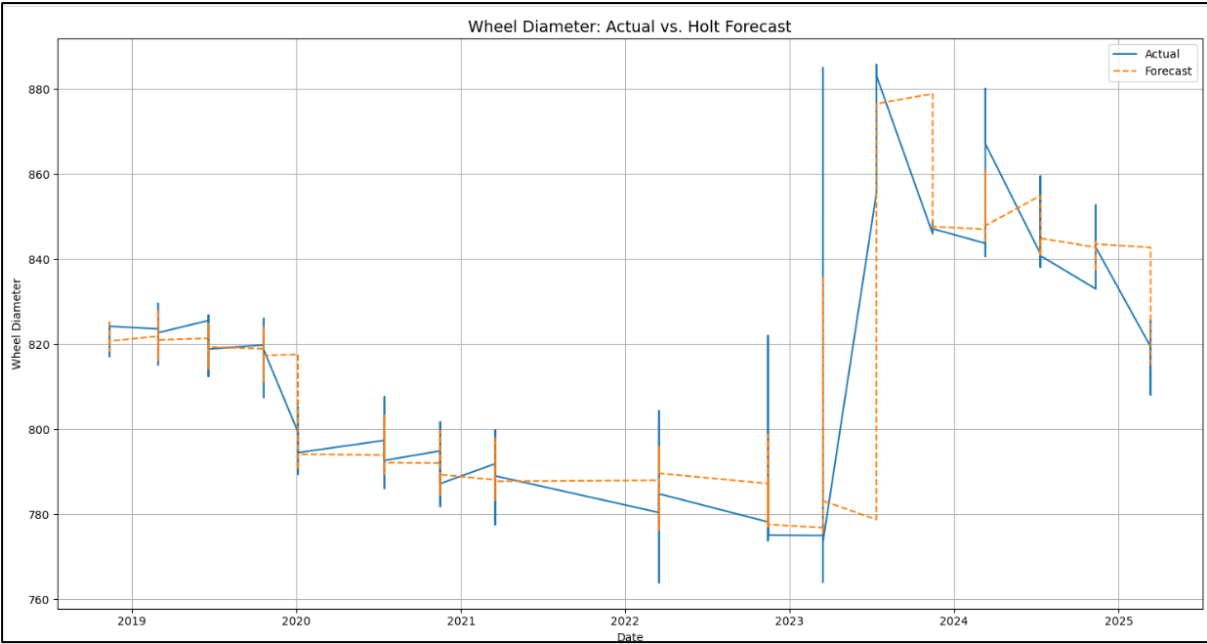
    # store prediction and actual value
    predictions2.append(yhat)
    actuals.append(testing_mtbef.iloc[i])

# add actual value to training data
training_mtbef = pd.concat([training_mtbef, pd.DataFrame([testing_mtbef.iloc[i]], columns=training_mtbef.columns)], ignore_index=True)

```

Figure 4.3. 5 One-Step ahead forecast of all target variables.

The Holt-Winters model's robustness and adaptability were assessed using a rolling forecasting strategy. With this method, one-step-ahead forecasts are generated repeatedly, with each forecasting the next point after refitting the model and adding the actual observed value to the training set as shown in Figure 4.3.5. This iterative procedure simulates a real-world situation in which maintenance systems are updated regularly in response to new data. By recalibrating the model after each step, the approach not only measures the predictive accuracy but also tests the model's ability to adapt to ongoing changes in the data. Compared to static testing methods, this rolling evaluation offers a more useful and trustworthy indicator of model performance, particularly for time series in operational environments.



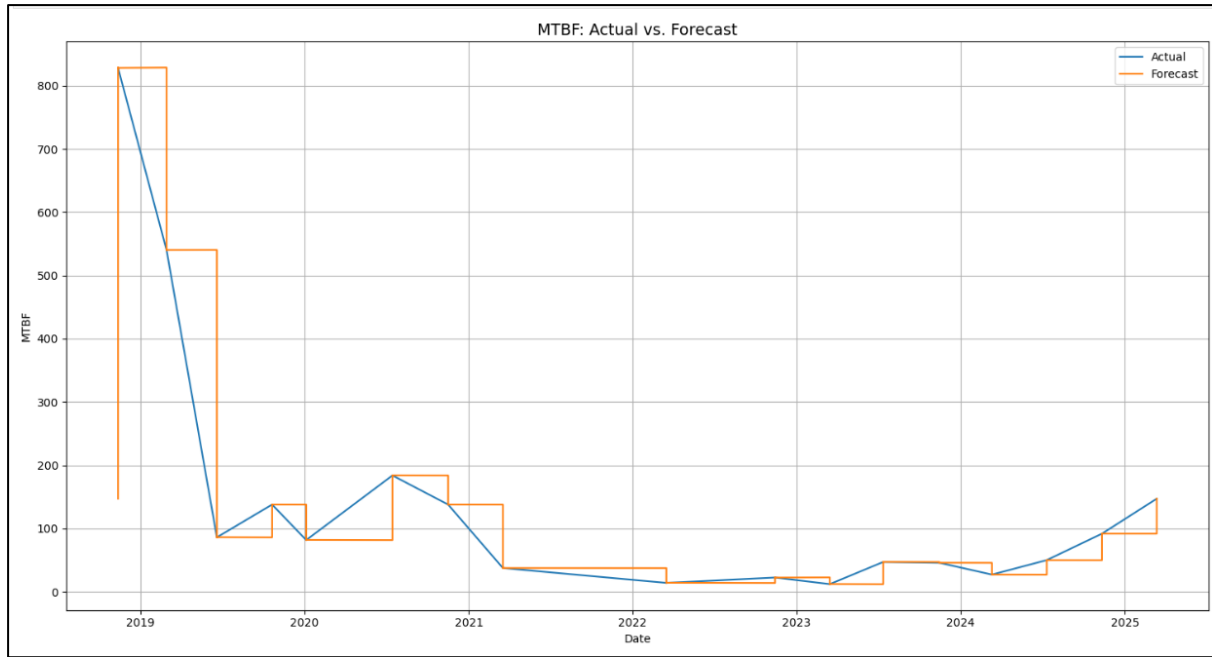


Figure 4.3. 6 Graph of actual and forecast of all target variables.

The predicted values for wheel diameter, wheel flange, and MTBF were plotted against the actual observed values in order to evaluate the forecasts' visual accuracy as shown in the Figure 4.3.6. Although slight underestimations were identified during sudden decreases, the model well represented the seasonal variations and the gradual downward trend for wheel diameter. The wheel flange predictions followed the actual values with reasonable accuracy, although slight deviations occurred, likely due to minor non-seasonal fluctuations not fully captured by the additive model. The model's projections accurately followed MTBF, a more unpredictable metric. The MTBF model's high smoothing level enabled it to quickly adapt to short-term changes, producing forecasts that closely reflected actual fluctuations. Overall, the visualisations demonstrated that the Holt-Winters approach could to identify and predict both periodic behaviours and long-term patterns across all three variables.

Metrics	Wheel Diameter	Wheel Flange	MTBF
R^2	0.7820	0.6808	0.9331
MSE	142.1211	0.5148	1597.2671
RMSE	11.9215	0.7175	39.9658
MAE	5.6349	0.5013	4.1458
MAPE (%)	0.69	1.70	3.65

Table 4.3.1 Evaluation of all target variables for Holt Winters Method.

Quantitative evaluation of the Holt-Winters model was performed using several statistical metrics, including R-squared (R^2), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) as shown in the Table 4.3.1. For the wheel diameter forecasts, the model achieved an R^2 score of 0.7820, indicating strong explanatory power. The MSE was 142.1211, the RMSE was 11.9215, the MAE was 5.6349, and the MAPE was remarkably low at 0.69%. For wheel flange, the R^2 score was 0.6808, showing moderate fit, with an MSE of 0.5148, RMSE of 0.7175, MAE of 0.5013, and MAPE of 1.70%. The MTBF forecasts yielded the highest performance, with an R^2 score of 0.9331, suggesting an excellent fit to the actual data. The model's MSE for MTBF was 1597.2671, the RMSE was 39.9658, the MAE was 4.1458, and the MAPE was 3.65%. These results validate the model's effectiveness in forecasting each variable, particularly MTBF, which exhibited a high degree of variance.

TrainID	Date	Forecast
0	T01 2026-03-31	811.348975
1	T01 2026-06-30	811.653313
2	T01 2026-09-30	811.233759
3	T01 2026-12-31	811.798696
4	T05 2026-03-31	811.348975
5	T05 2026-06-30	811.653313
6	T05 2026-09-30	811.233759
7	T05 2026-12-31	811.798696
8	T25 2026-03-31	811.348975
9	T25 2026-06-30	811.653313
10	T25 2026-09-30	811.233759
11	T25 2026-12-31	811.798696
12	T45 2026-03-31	811.348975
13	T45 2026-06-30	811.653313
14	T45 2026-09-30	811.233759
15	T45 2026-12-31	811.798696
16	T46 2026-03-31	811.348975
17	T46 2026-06-30	811.653313
18	T46 2026-09-30	811.233759
19	T46 2026-12-31	811.798696
20	T55 2026-03-31	811.348975
21	T55 2026-06-30	811.653313
22	T55 2026-09-30	811.233759
23	T55 2026-12-31	811.798696
24	T57 2026-03-31	811.348975
25	T57 2026-06-30	811.653313
26	T57 2026-09-30	811.233759
27	T57 2026-12-31	811.798696
28	T58 2026-03-31	811.348975
29	T58 2026-06-30	811.653313
30	T58 2026-09-30	811.233759
31	T58 2026-12-31	811.798696

Forecast for
Wheel Diameter

TrainID	Date	Forecast
0	T01 2026-03-31	30.148552
1	T01 2026-06-30	29.993013
2	T01 2026-09-30	29.917007
3	T01 2026-12-31	30.114324
4	T05 2026-03-31	30.148552
5	T05 2026-06-30	29.993013
6	T05 2026-09-30	29.917007
7	T05 2026-12-31	30.114324
8	T25 2026-03-31	30.148552
9	T25 2026-06-30	29.993013
10	T25 2026-09-30	29.917007
11	T25 2026-12-31	30.114324
12	T45 2026-03-31	30.148552
13	T45 2026-06-30	29.993013
14	T45 2026-09-30	29.917007
15	T45 2026-12-31	30.114324
16	T46 2026-03-31	30.148552
17	T46 2026-06-30	29.993013
18	T46 2026-09-30	29.917007
19	T46 2026-12-31	30.114324
20	T55 2026-03-31	30.148552
21	T55 2026-06-30	29.993013
22	T55 2026-09-30	29.917007
23	T55 2026-12-31	30.114324
24	T57 2026-03-31	30.148552
25	T57 2026-06-30	29.993013
26	T57 2026-09-30	29.917007
27	T57 2026-12-31	30.114324
28	T58 2026-03-31	30.148552
29	T58 2026-06-30	29.993013
30	T58 2026-09-30	29.917007
31	T58 2026-12-31	30.114324

Forecast for
Wheel Flange

TrainID	Date	Forecast
0	T01 2026-03-31	147.262886
1	T01 2026-06-30	147.442981
2	T01 2026-09-30	147.344513
3	T01 2026-12-31	147.387407
4	T05 2026-03-31	147.262886
5	T05 2026-06-30	147.442981
6	T05 2026-09-30	147.344513
7	T05 2026-12-31	147.387407
8	T25 2026-03-31	147.262886
9	T25 2026-06-30	147.442981
10	T25 2026-09-30	147.344513
11	T25 2026-12-31	147.387407
12	T45 2026-03-31	147.262886
13	T45 2026-06-30	147.442981
14	T45 2026-09-30	147.344513
15	T45 2026-12-31	147.387407
16	T46 2026-03-31	147.262886
17	T46 2026-06-30	147.442981
18	T46 2026-09-30	147.344513
19	T46 2026-12-31	147.387407
20	T55 2026-03-31	147.262886
21	T55 2026-06-30	147.442981
22	T55 2026-09-30	147.344513
23	T55 2026-12-31	147.387407
24	T57 2026-03-31	147.262886
25	T57 2026-06-30	147.442981
26	T57 2026-09-30	147.344513
27	T57 2026-12-31	147.387407
28	T58 2026-03-31	147.262886
29	T58 2026-06-30	147.442981
30	T58 2026-09-30	147.344513
31	T58 2026-12-31	147.387407

Forecast for
MTBF

Figure 4.3.7 Forecast of all target variables.

After evaluation of the model, it was used to predict wheel diameter, wheel flange, and MTBF values for the upcoming four quarters as shown in the Figure 4.3.7. The forecasts that are produced provide a perspective on the expected decrease and maintenance requirements for

train components. The model predicted a continuous decrease in wheel diameter that was in line with long-term wear patterns. While MTBF forecasts indicated possible fluctuations that could be essential for predicting system dependability, wheel flange forecasts indicated minor changes. Engineers and maintenance planners can use these forecasts to assist them proactively schedule maintenance and inspections.

4.4 Deep Learning

```
[ ] test_size = 3938
    test_ind = len(df) - test_size

[ ] train = df.iloc[:test_ind]
    test = df.iloc[test_ind:]
```

Figure 4.4. 1: Test and Train data

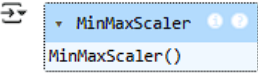
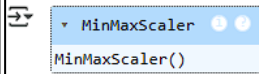
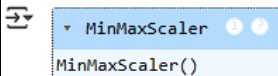
<pre>[] #Feature scaling from sklearn.preprocessing import MinMaxScaler numerical_cols = ['WheelFlange'] scaler = MinMaxScaler() scaler.fit(train[numerical_cols])</pre>	<pre>[] #Feature scaling from sklearn.preprocessing import MinMaxScaler numerical_cols = ['WheelDiameter'] scaler = MinMaxScaler() scaler.fit(train[numerical_cols])</pre>
	
<pre>[] #Feature scaling from sklearn.preprocessing import MinMaxScaler numerical_cols = ['MTBF'] scaler = MinMaxScaler() scaler.fit(train[numerical_cols])</pre>	
	

Figure 4.4. 2: Feature Scaling

First, the dataset is split into training and testing sets by reserving the last 3,938 records as the test set, while the remaining earlier records are used for training as shown in Figure 4.4.1. This method preserves the chronological order of the data, which is crucial for forecasting tasks to simulate real-world scenarios where past data is used to predict future outcomes. Following the split, feature scaling is applied using 'MinMaxScaler' as shown in Figure 4.4.2. Before training the GRU model, the input features were scaled using MinMaxScaler, which transforms

the values of each feature to a fixed range, typically between 0 and 1. This step is important because GRU work better when all input values are within a similar range. Without scaling, features with large values could dominate the training process and make learning difficult or unstable. The scaler was trained using only the training data to avoid using future information, which would cause data leakage. After this, the same scaling was applied to both training and testing data. This careful approach helps the model train more effectively and ensures fair performance evaluation.

```

from keras.layers import Dropout
from keras.optimizers import Adam

gru_model = Sequential()

# Define the number of features
n_features = 1 # Get the number of columns from the scaled training data

gru_model.add(GRU(50, activation='relu', input_shape=(length, n_features)))
gru_model.add(Dropout(0.2))

gru_model.add(Dense(1))

optimizer = Adam(learning_rate=0.0005) #default is 0.001;try lowering it to 0.0005
gru_model.compile(optimizer=optimizer, loss='mse')

gru_model.summary()

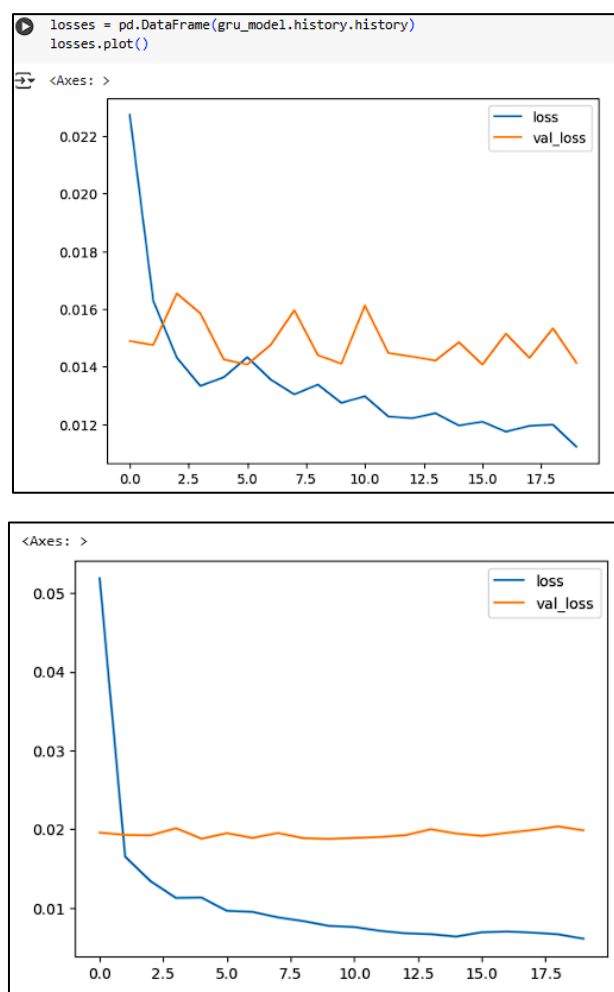
```

Model: "sequential"		
Layer (type)	Output Shape	Param #
gru (GRU)	(None, 50)	7,950
dropout (Dropout)	(None, 50)	0
dense (Dense)	(None, 1)	51
Total params: 8,001 (31.25 KB)		
Trainable params: 8,001 (31.25 KB)		
Non-trainable params: 0 (0.00 B)		

Figure 4.4. 3: Developing GRU Model

We develop GRU model as shown in Figure 4.4.3. It begins by importing the essential components, including the Dropout layer and the Adam optimizer. The model, built using the Sequential API, has one GRU layer with 50 units with ReLU activation. It is designed to handle sequential input of shape (length, n_features), where length refers to the number of time steps

and `n_features` is set to 1, indicating a univariate input. This is followed by a 20% dropout layer to reduce overfitting. For regression tasks, a final Dense layer with only one output unit is introduced. The Adam optimizer is used to construct the model, with a 0.0005 learning rate and a Mean Squared Error (MSE) loss function. The model summary shows a total of 8,001 trainable parameters: 7,950 from the GRU layer and 51 from the Dense layer. This simple architecture is suited for basic sequence prediction tasks but might be inadequate for highly volatile data like WheelFlange, WheelDiameter, or MTBF if only a single historical variable is used for forecasting.



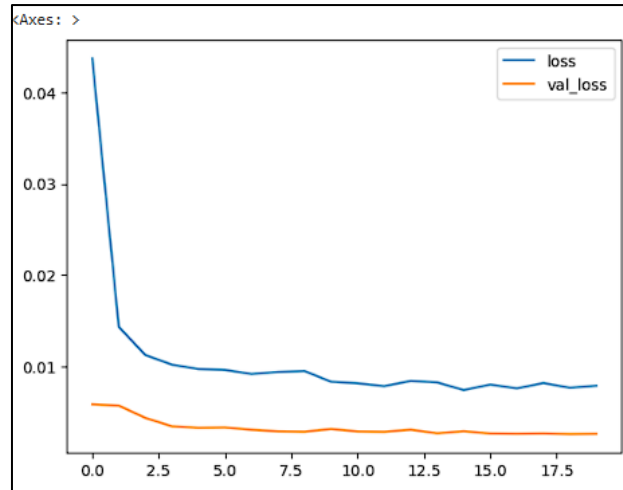
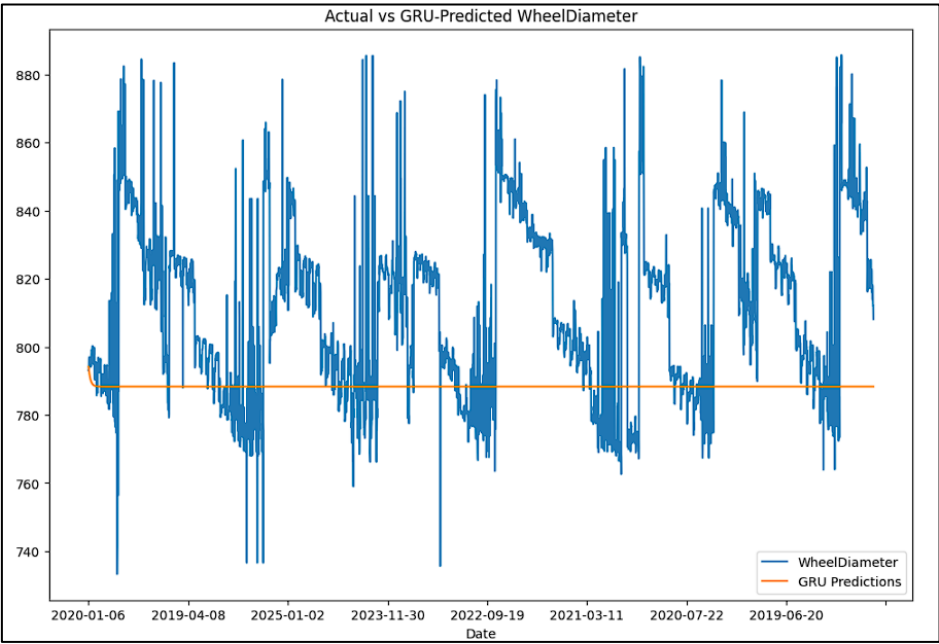
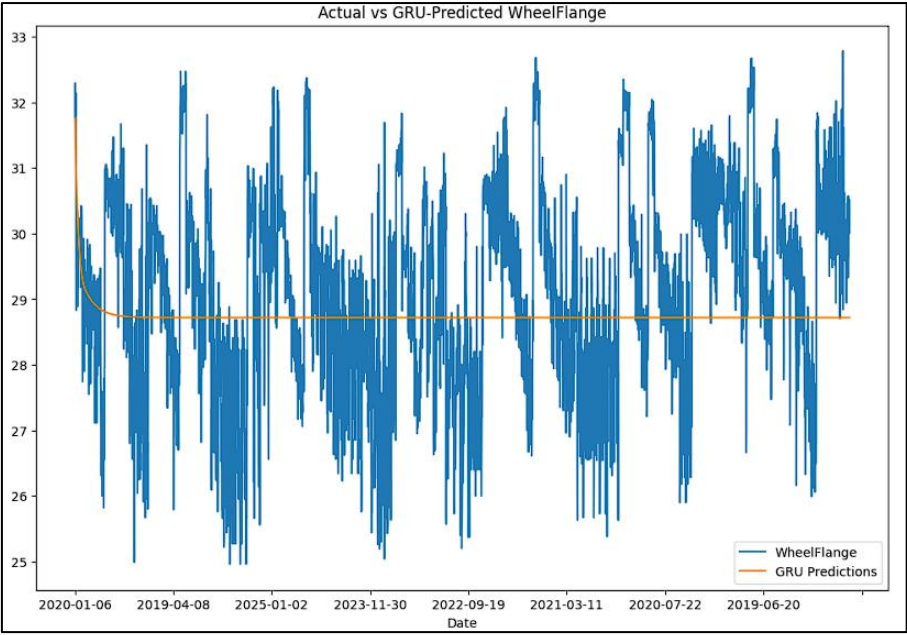


Figure 4.4. 4: Training and Validation Loss Plot for Wheel Flange, Wheel Diameter and MTBF

The plots illustrate in Figure 4.4.4 represent training and validation loss curves for GRU deep learning models predicting Wheel Flange, Wheel Diameter, and MTBF. It offers valuable insights into the models' learning behaviours. In the first plot, both training and validation losses initially decrease, but the validation loss remain flat and fluctuates after several epochs, suggesting a tendency toward overfitting. The widening gap between training and validation loss further supports this observation, indicating that while the model fits the training data well, it fails to generalize effectively. The second plot reveals a more severe case of overfitting. Although the training loss decreases sharply and continues to improve, the validation loss quickly remains flat and remains stagnant. It is implying that the model memorizes the training data but performs poorly on unseen data. This issue necessitates the implementation of strategies such as early stopping, increased regularization, or reduced model complexity. The third plot presents a peculiar scenario where the validation loss remains consistently lower than the training loss, which is highly unusual and may indicate data leakage or inconsistencies in data preprocessing. Potential causes include improper data splitting, misuse of scaling techniques, or a validation set that is not representative of the real data distribution. Although this plot shows the lowest validation loss among the three, the model's apparent performance is likely misleading unless these issues are resolved. Overall, these observations underscore the importance of careful validation and regularization to ensure reliable and generalizable predictive performance in deep learning models.



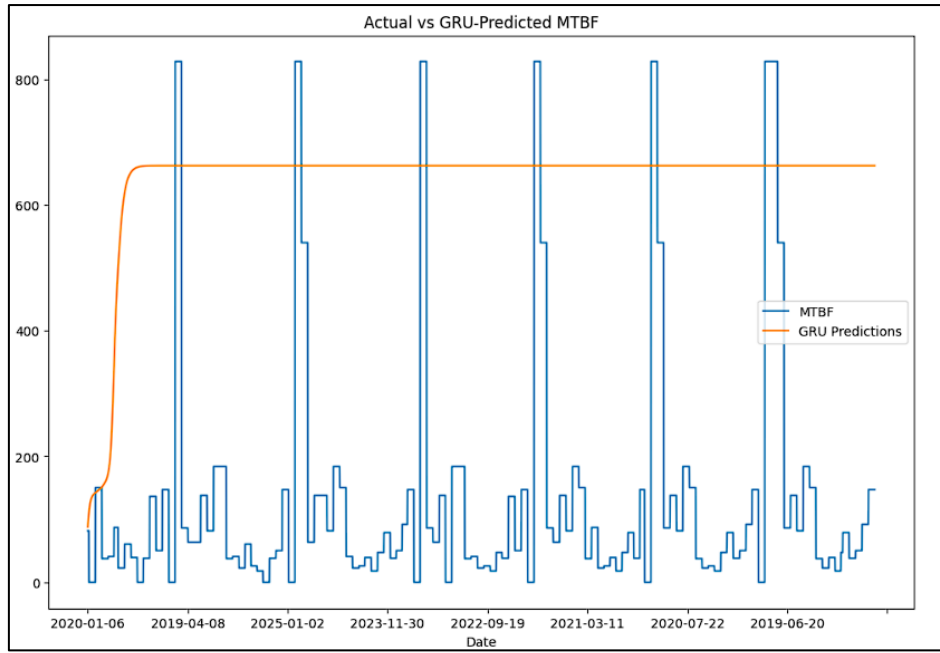


Figure 4.4.5: Actual vs GRU Prediction of all Target Variables

The Actual vs. GRU-Predicted plots in Figure 4.4.5 for Wheel Flange, Wheel Diameter, and MTBF reveal that the GRU models perform poorly in forecasting all three variables. For Wheel Flange and Wheel Diameter, the actual values show high volatility with rapid changes, but the GRU predictions are almost flat lines, failing to capture any real patterns. This indicates severe underfitting, where the model is unable to learn or represent the data trends. These results support the overfitting issues seen earlier in the training and validation loss plots, where the model performs well on training data but poorly on new data. For MTBF, the actual values appear step-like with sudden jumps, but the GRU model predicts a constant high value, completely missing the step pattern. This suggests the model is not suitable for this kind of data, which might need a different approach such as event-based or survival analysis models. Although the validation loss plot shows overfitting (low training loss, high validation loss), the actual predictions reveal underfitting behaviour. The model outputs nearly flat lines that do not follow the real patterns, suggesting it failed to learn the time-based changes. This mismatch indicates that the model memorized training data without learning generalizable features, leading to poor forecast performance and severe model limitations. Overall, the GRU models act like constant predictors and fail to provide useful forecasts, highlighting major issues in the model setup and the need for better model selection or data preprocessing strategies.

Metrics	Wheel Diameter	Wheel Flange	MTBF
R^2	-0.7512	-984.8790	-6.904
MSE	1084.9745	610821.1157	307622.8401
RMSE	32.9390	781.5505	554.6376
MAE	25.9673	781.1536	531.7551
MAPE (%)	3.14%	96.45%	inf%

Table 4.4. 1: Evaluation of all Target Variables for GRU Model

The performance evaluation in Table 4.4.1 of the GRU model reveals significant limitations in its ability to accurately forecast Wheel Diameter, Wheel Flange, and MTBF for Wheel Diameter, the R^2 value of -0.7512 indicates that the model performs worse than a simple average-based predictor, suggesting it fails to capture the variance in the data. Although the Mean Absolute Percentage Error (MAPE) is relatively low at 3.14%, this may be misleading due to large absolute errors ($MAE = 25.97$), which imply the model lacks precision. In the case of Wheel Flange, the model exhibits extremely poor predictive performance, with an R^2 of -984.8790 and a MAPE of 96.45%, indicating predictions that are nearly 100% inaccurate on average. This likely results from the periodic reset patterns in the flange data, possibly due to maintenance activities that the model fails to account for. Similarly, MTBF prediction is weak, with a negative R^2 value (-6.9040) and an infinite MAPE, which is often caused by actual MTBF values of zero in the test set—leading to undefined percentage errors.

4.5 Forecasting Using Best Model

This section provides the forecasting results using the top-performing model that was chosen following a comparative analysis. Holt-Winters was selected as the preferred model between the Holt-Winters Exponential Smoothing method and the GRU neural network model. Its better accuracy to identify seasonality and trend components, reduced forecasting error, and improved interpretability within the framework of our time series data served as the foundation for our choice. The Holt-Winters model forecasts for wheel diameter, wheel flange, and MTBF are described in the following.

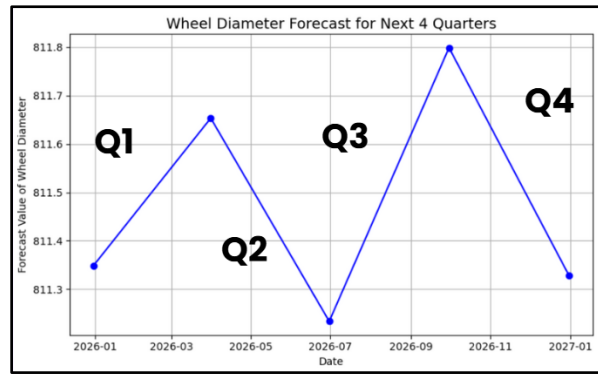


Figure 4.5.1 Wheel Diameter Forecast.

Over the period of the four quarters, the Wheel Diameter Forecast graph in Figure 4.5.1 displays rather small changes as shown in the Figure 4.5.1. The diameter starts around 811.3 mm in Q1 (2026-01) and increases to around 811.7 mm in Q2 (2026-04). After a decrease to almost 811.25 mm in Q3 (2026-07), it rises at over 811.8 mm in Q4 (2026-10) and then decreases a little more until the end of the year. Overall, the diameter maintains within a limitation on range, indicating secure wheel conditions, although these minor variations show regular wear and maintenance cycles.

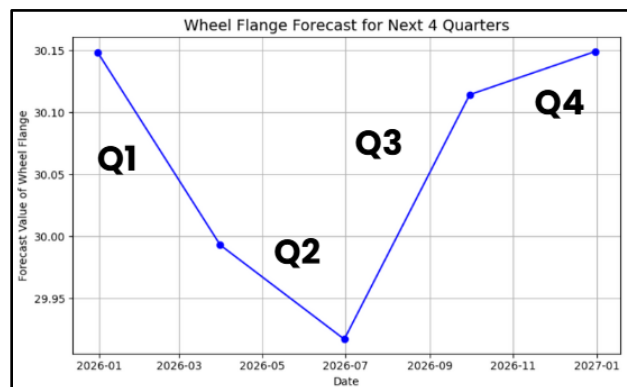


Figure 4.5. 2 Wheel Flange Forecast

The Wheel Flange Forecast graph in Figure 4.5.2 reveals a U-shaped trend across the quarters as shown in the Figure 4.5.2. In Q1 (2026-01), it rises at about 30.15 mm, then decreases over time through Q2 (2026-04), and reaches its lowest point around 29.95 mm in Q3 (2026-07). In Q4 (2026-10), the flange thickness returns to its initial value of around 30.15 mm. This pattern clearly indicates breakdown over time, followed by repairs such as wheel replacement or wheel profiling that restore the train to its previous state.

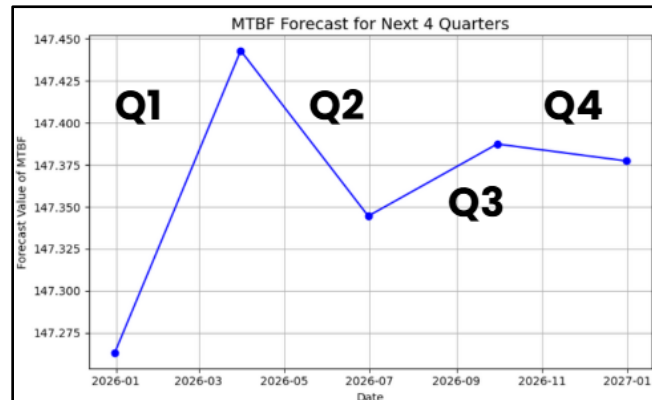


Figure 4.5 3 MTBF Forecast

The MTBF Forecast graph in Figure 4.5.3 demonstrates a pattern of improvement followed by slight decline and stabilization as shown in the Figure 4.5.3. Better system dependability is shown by MTBF, which starts at about 147,275 hours in Q1 (2026-01) and rises at about 147,450 hours in Q2 (2026-04). But in Q3 (2026-07), it decreases to about 147,350 hours, and then slightly recovers to 147,375 hours in Q4 (2026-10). The effects of maintenance or upgrades, operating stress, and corrective actions to stabilise reliability may be reflected in these patterns.

4.6 Graphical User Interface

A wheel maintenance prediction system is the purpose of this graphical user interface as shown in the Figure 4.6.1. Users can input key wheel parameters and examine the status of their maintenance. For more efficient decision-making, the system offers failure analysis, maintenance schedules, real-time status, and future predictions.

Wheel Maintenance Prediction System

Input Parameters
Train ID: (Available train data)
Car: (Motor/Trailer car)
Bogie: (Bogie selection)
Axle: (Axle position)
Date: (YYYY-MM-DD)
Wheel Diameter (mm): (Replacement if < 775mm)
Wheel Flange (mm): (Profiling if < 27.5mm)
Failures per Quarter: (For MTBF calculation)

Prediction Results

☒ Current Status
☒ Predictions
☐ Schedule
☐ Analysis

WHEEL MAINTENANCE PREDICTION SYSTEM

Welcome to the Wheel Maintenance Prediction System!

MAINTENANCE CRITERIA:

- Wheel Replacement: Required when diameter < 775mm
- Wheel Profiling: Required when flange < 27.5mm
- Failure: Occurs when either condition is met

MTBF CALCULATION:
MTBF = (18 x Days in Quarter) / Failures per Quarter

INSTRUCTIONS:

- Fill in all input parameters on the left panel
- Click 'Predict Maintenance' to generate analysis
- Review results in the organized tabs:
 - Current Status - Current wheel condition
 - Predictions - Future predictions
 - Schedule - Maintenance timeline
 - Analysis - Detailed MTBF analysis

Ready for prediction!

Figure 4.6. 1 GUI of wheel prediction system

The system offers a comprehensive interface for wheel maintenance analysis and prediction. Users will get clear instructions on how to enter the necessary input settings as soon as the GUI opens. These include the Train ID, Car, Bogie, Axle, Date, Wheel Diameter (mm), Wheel Flange (mm), and Failure per Quarter. Once the necessary data is filled in, clicking the “Predict Maintenance” button will display the Current Status section.

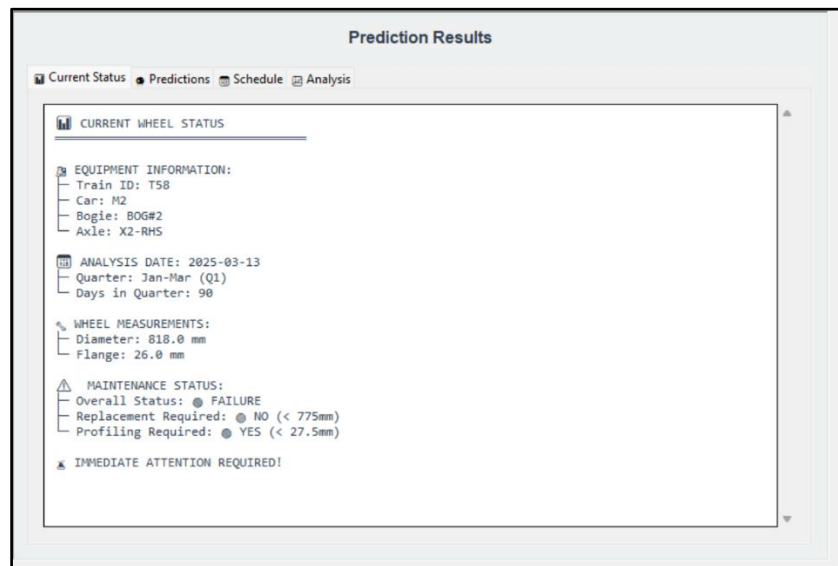


Figure 4.6. 2 Current Status Button

According to the user's input, the "Current Status" button displays the most recent maintenance overview as shown in Figure 4.6.2. First, it displays the equipment information just as the user entered them. The analysis date and the most current wheel measurements are then displayed. The maintenance status is provided as well in this part, along with the overall status, showing if a failure has been discovered or not, whether replacement is necessary, and whether wheel profiling is required. Users may see the wheel's current condition right away due to this.

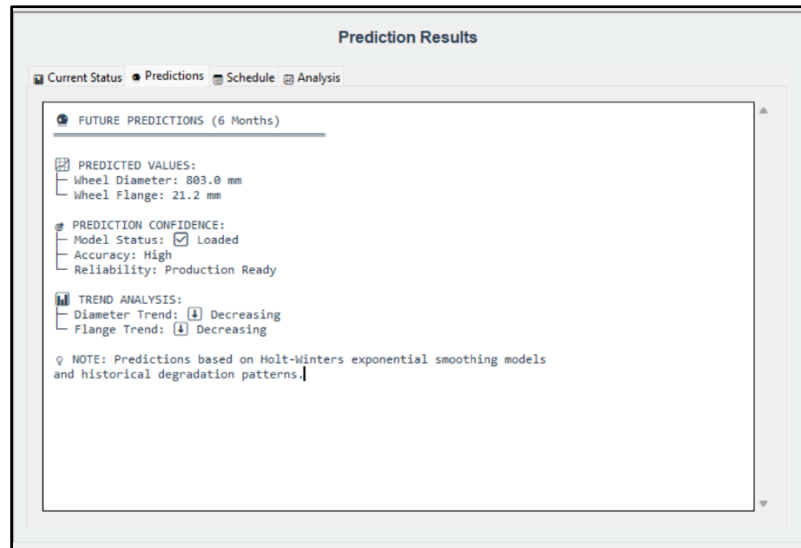


Figure 4.6. 3 Predictions Button

According to Figure 4.6.3 users can check future projections and forecast wheel condition six months in advance by clicking the "Predictions" button. It shows the expected values for the flange and wheel diameter. Prediction accuracy (high or poor), prediction confidence (showing whether the model is loaded or not), and dependability status (showing whether the model is production-ready) also feature in this section.

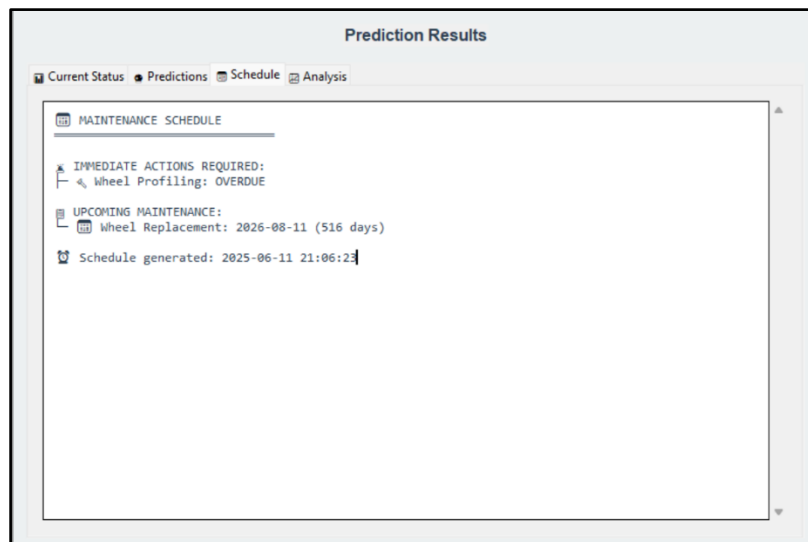


Figure 4.6. 4 Schedule Button

By clicking the "Schedule" button, users can access the upcoming maintenance schedule as shown in Figure 4.6.4. This includes any necessary quick fixes, including determining whether wheel profiling is past due. Additionally, it also shows the upcoming maintenance schedule, which specifies when the next wheel replacement is expected to occur.

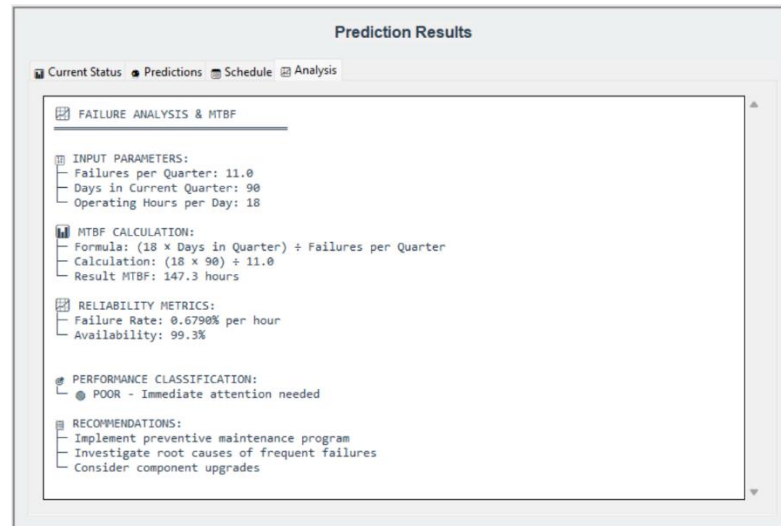


Figure 4.6. 5 Analysis Button

The "Analysis" button offers detailed data on MTBF (Mean Time Between Failures) and failure analysis as shown in Figure 4.6.5. The MTBF is calculated and shown using the quarterly failure input value. Additionally, the system offers reliability metrics such as availability, which is calculated as $[1 - \text{Failure} / (18 \times \text{Days})] \times 100$, which shows the percentage of time the wheel remains in optimal condition, and failure rate, which is calculated as $(1/\text{MTBF}) \times 100$, which represents the frequency of failures per hour. Additionally, it provides maintenance teams with useful guidance and classifies performance (e.g., if it is poor or not).

Last but not least, the maintenance staff can easily reset the system and conduct a fresh investigation by clicking the "Clear Results" button if they want to analyse another wheel.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

Three unique analytical methods were used in this research to enhance railway maintenance forecasting and classification: Holt-Winters Method, GRU and LDA. Each technique was chosen for its distinct characteristics in managing time-series or categorical data related to critical maintenance indicators such as wheel diameter, wheel flange, and MTBF. The goal of this study was to provide a thorough understanding of maintenance trends and failure risks by studying the data from various methodology viewpoints.

The Holt-Winters technique was useful for studying components with distinct seasonal patterns and steady trends, such as wheel diameter and flange. Its capacity to model level, trend, and seasonality contributed to consistent and interpretable forecasts. In contrast, the GRU model, a deep learning technique well-suited to sequential data, performed well in forecasting MTBF. Its capacity to capture nonlinear interactions and learn from long-term dependencies made it appropriate for more dynamic and complicated information, where traditional models may fail.

In addition to forecasting, LDA was utilized to identify and distinguish between different maintenance conditions. LDA assisted in identifying patterns that differed between failure and non-failure scenarios, providing vital insights for preventive action. Overall, by employing all three models independently, the study was able to offer focused findings that assist better maintenance planning and decision-making, resulting in safer and more efficient railway operations.

5.2 Limitations

Despite the promising outcomes of the Wheel Maintenance Prediction System, several limitations were encountered during the development process. The limited time for model tuning, particularly for the Holt-Winters and GRU models, was a major limitation. Extended validation and ideal hyperparameter tuning could not be completed due to time restrictions.

The irregularity of data intervals presented another difficulty. Wheel measurements should ideally be taken on a quarterly basis, however several trains had inconsistent or missing records, which made it difficult for the model to accurately identify temporal patterns. Another limitation was the small sample size. Due to computational constraints and the time needed to train the models, only data from eight of the 58 obtainable trains were used. The variability in real-world situations might not be accurately reflected by the small sample.

Furthermore, the range of features that were accessible was constrained. Wheel flange, wheel diameter, and MTBF were the only features used in the model's development. Important factors that could have improved the model's contextual knowledge and prediction performance, such track condition, weather, or load patterns, were not offered.

5.3 Recommendations

A few of recommendations are made to strengthen the system's reliability and applicability. First, adding real-time monitoring to the system is recommended. Based on current wheel conditions, this would enable more precise and immediate maintenance planning. Second, to make sure maintenance employees know how to evaluate the predictive outputs, training sessions should be held. Teams will be able to use the GUI for proactive planning and decision-making with confidence as an outcome.

Additionally, for improved feature engineering, collaboration with domain experts is strongly encouraged. The model can be made more accurate and useful in practice by adding additional relevant variables, such as track conditions or environmental influences. Finally, the system should be modified to include a greater number of trains as time and computational resources allow, guaranteeing more accurate and general predictions throughout the entire fleet of trains.

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